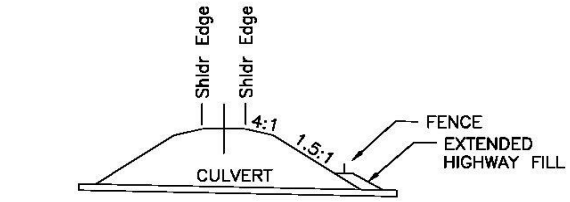
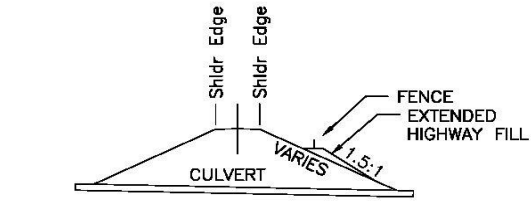


PROFILE
A–A
(ALONG FENCE LINE)



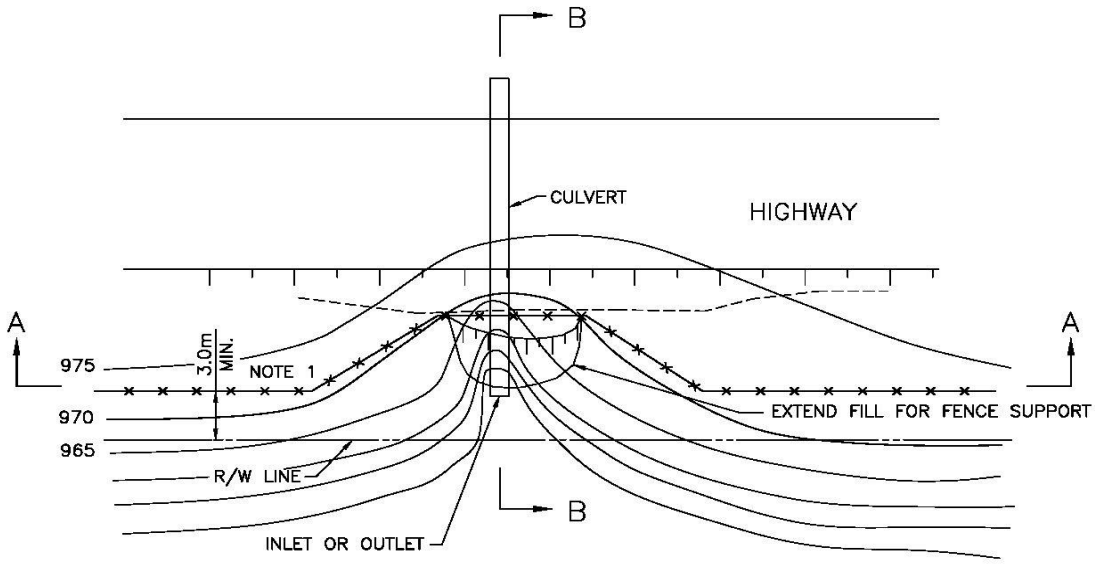
INSTALLATION ON NEW HIGHWAY CONSTRUCTION

SECTION
B–B



INSTALLATION ON EXISTING HIGHWAY FILL SLOPES

SECTION
B–B



PLAN

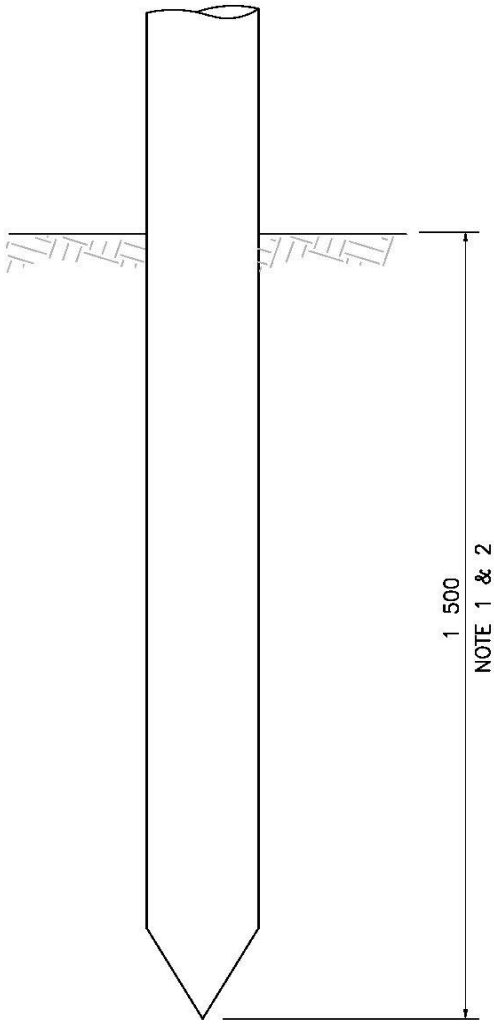
FENCE TREATMENT AT LARGE GULLIES

NOTES:
1. FENCE TO BE CONSTRUCTED INSIDE THE RIGHT–OF–WAY LINE AND MUST BE OUTSIDE OF THE HIGHWAY CLEAR ZONE (9.0m MINIMUM).
2. LOCKABLE HUMAN ACCESS GATE MUST BE PROVIDED WITHIN 25m OF CULVERTS THAT ARE GREATER THAN 2m IN DIAMETER.

BRITISH COLUMBIA — Ministry of Transportation & Infrastructure	WILDLIFE EXCLUSION FENCING FENCE TREATMENTS AT LARGE GULLIES	
No. / Revision / Date	Approved Issue Date:	SPECIFICATION DRAWING No. SP700–05

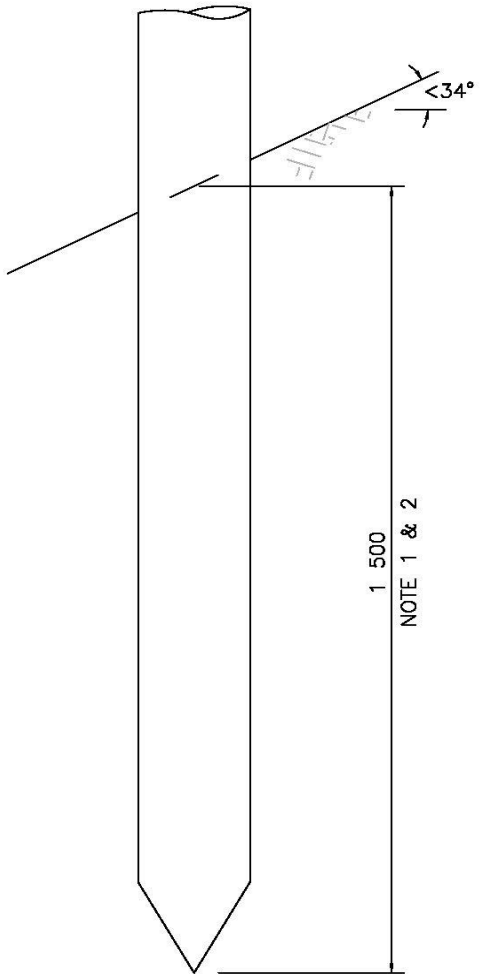
LEVEL GROUND

* ALL POSTS



SLOPES LESS THAN 34°

* ALL POSTS



FENCING IN EARTH

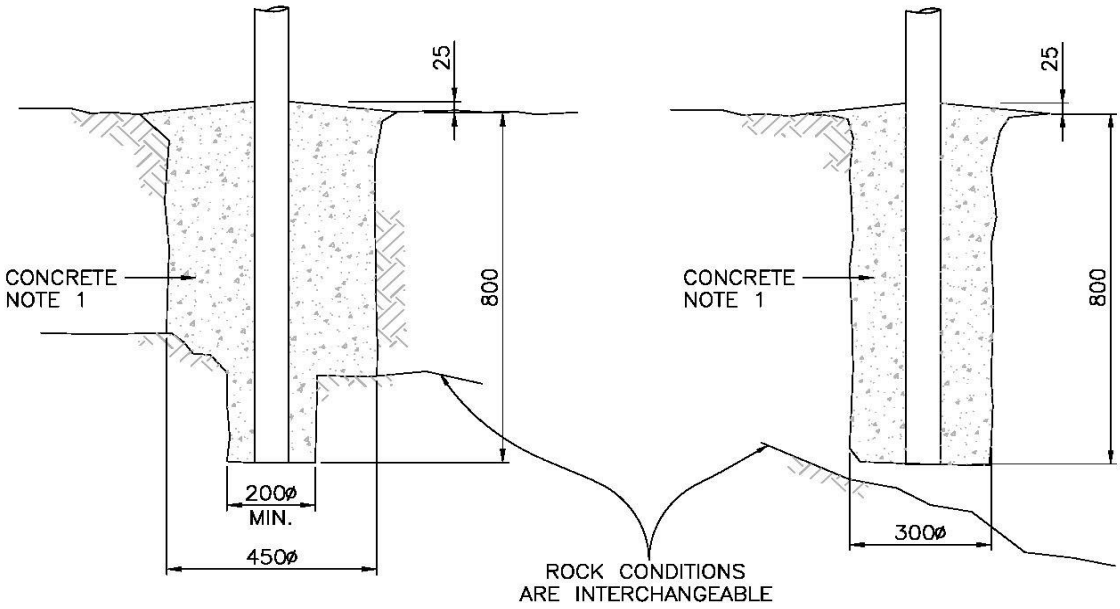
(WOOD POSTS)

1. 1500mm MINIMUM PROVIDED THAT IT MEETS THE REQUIREMENTS OF THE HORIZONTAL LOAD TEST.
2. WHEREVER THE REQUIREMENTS OF THE HORIZONTAL LOAD TEST CANNOT BE MET FOR WOOD POSTS, STEEL POSTS SHALL BE USED WHEN DIRECTED BY THE MINISTRY REPRESENTATIVE.

BRITISH COLUMBIA / Ministry of Transportation & Infrastructure	WILDLIFE EXCLUSION FENCING EMBEDMENT DETAILS FOR WOOD POSTS IN EARTH	
No. / Revision / Date	Approved Issue Date: JULY 2008	SPECIFICATION DRAWING No. SP700–06

- CORNER POSTS
- END POSTS

- LINE POSTS
- BRACE POSTS



GALVANIZED STEEL POSTS IN ORGANIC MATERIAL OR COMBINATION OF ROCK AND ORGANIC MATERIAL

EMBEDMENT DETAILS
NOT TO SCALE

NOTES:
1. CONCRETE SHALL CONFORM TO SS 218, CLASS C (20mm COARSE AGGREGATE).

BRITISH COLUMBIA | Ministry of Transportation & Infrastructure			WILDLIFE EXCLUSION FENCING EMBEDMENT DETAILS FOR FENCING IN EARTH	
No.	Revision	Date	Approved Issue Date:	SPECIFICATION DRAWING No. SP700–07

LEVEL GROUND

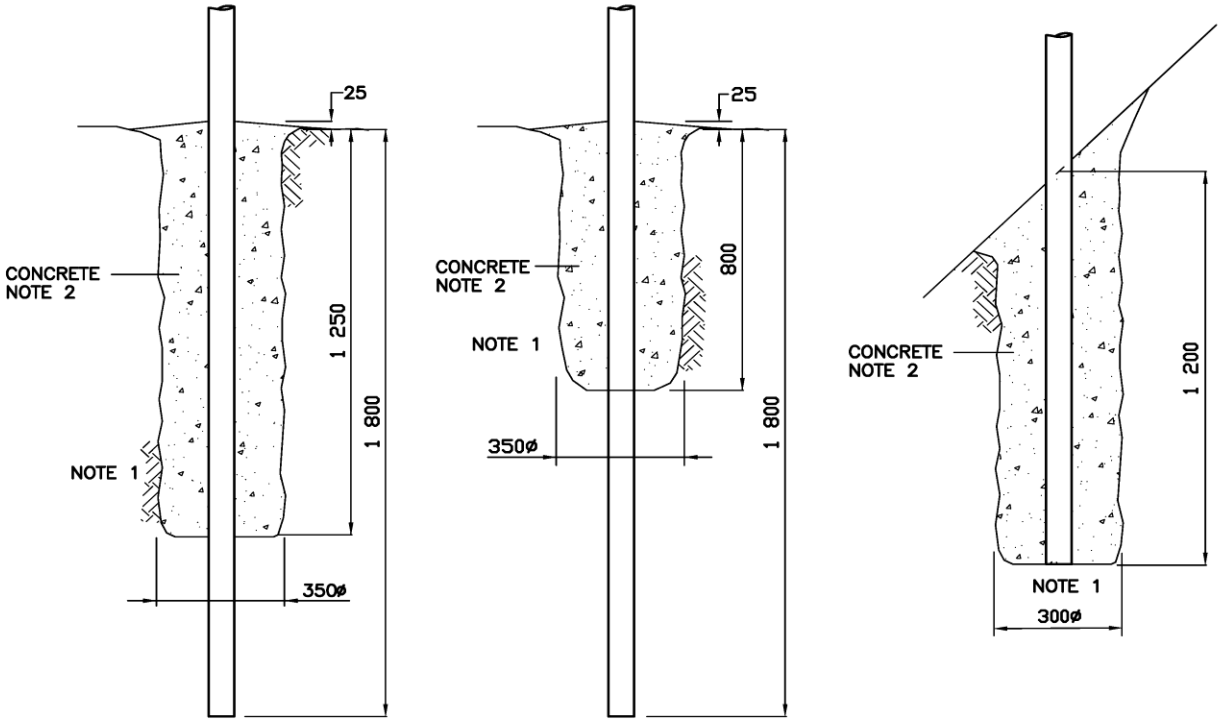
- CORNER POSTS
- END POSTS

LEVEL GROUND

- LINE POSTS
- BRACE POSTS

SLOPES
34° TO 50°

- ALL POSTS



FENCING IN SOFT ORGANIC SOILS OR WETLANDS (AS DIRECTED BY MINISTRY REPRESENTATIVE)

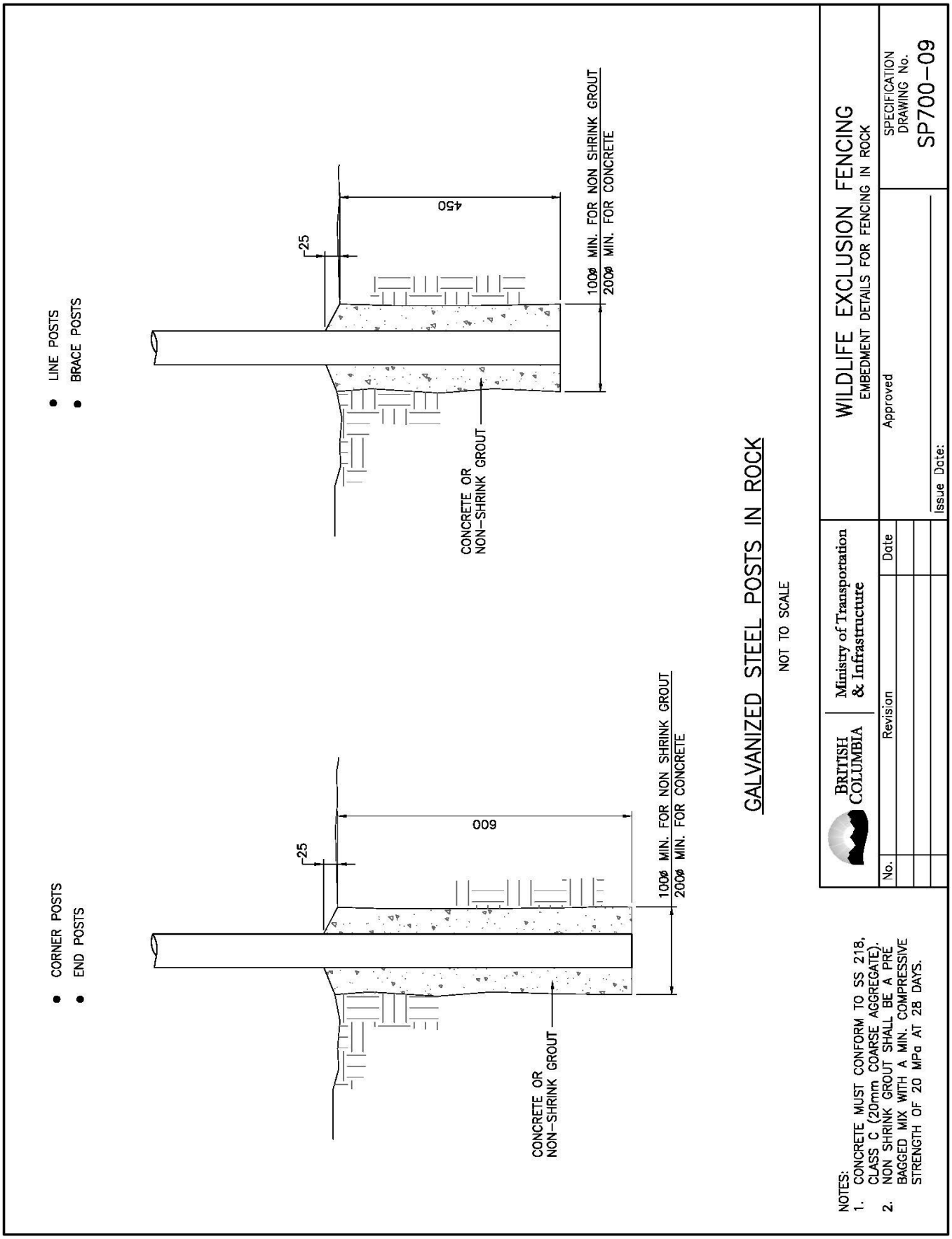
GALVANIZED STEEL POSTS

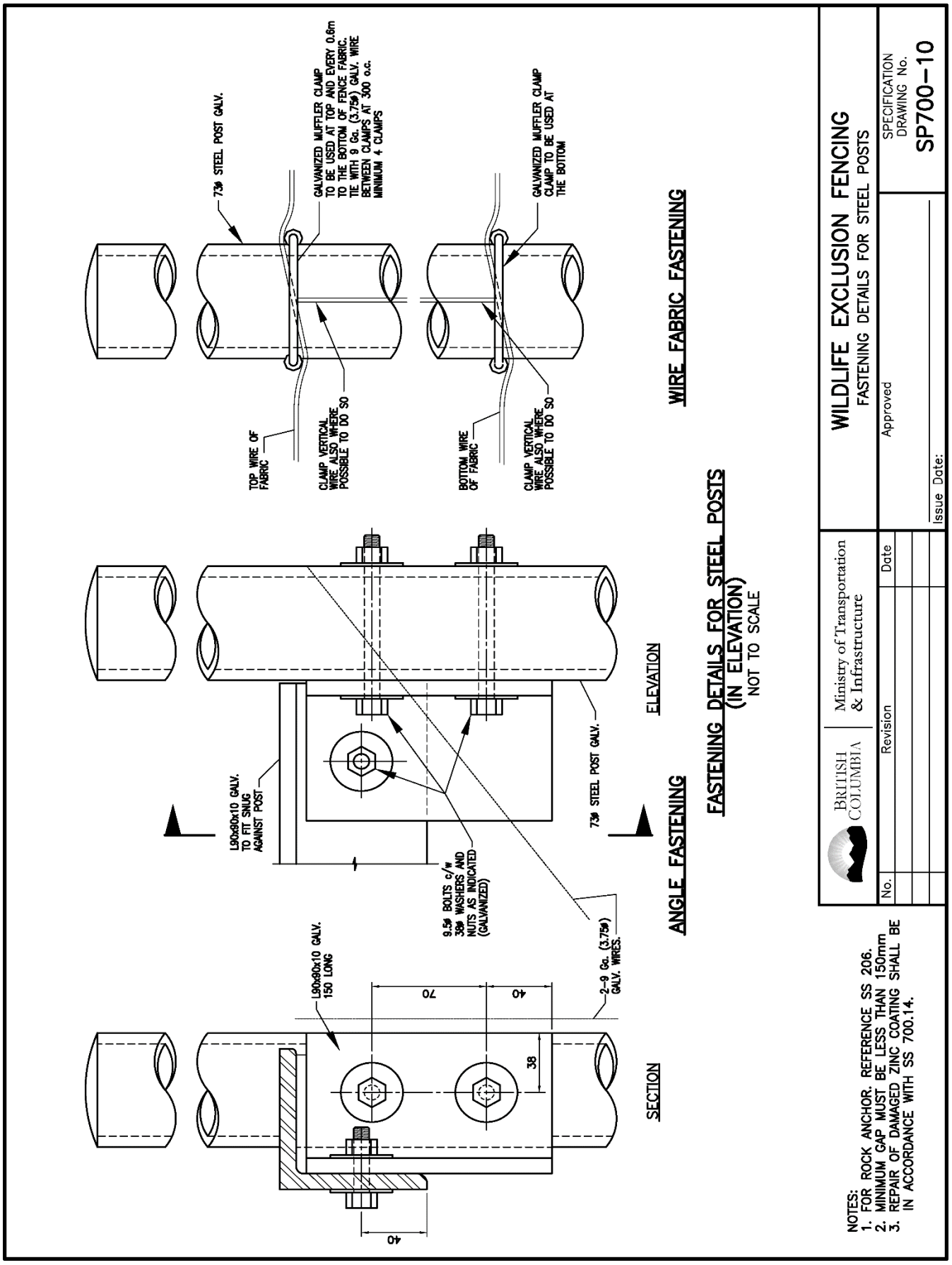
EMBEDMENT DETAILS

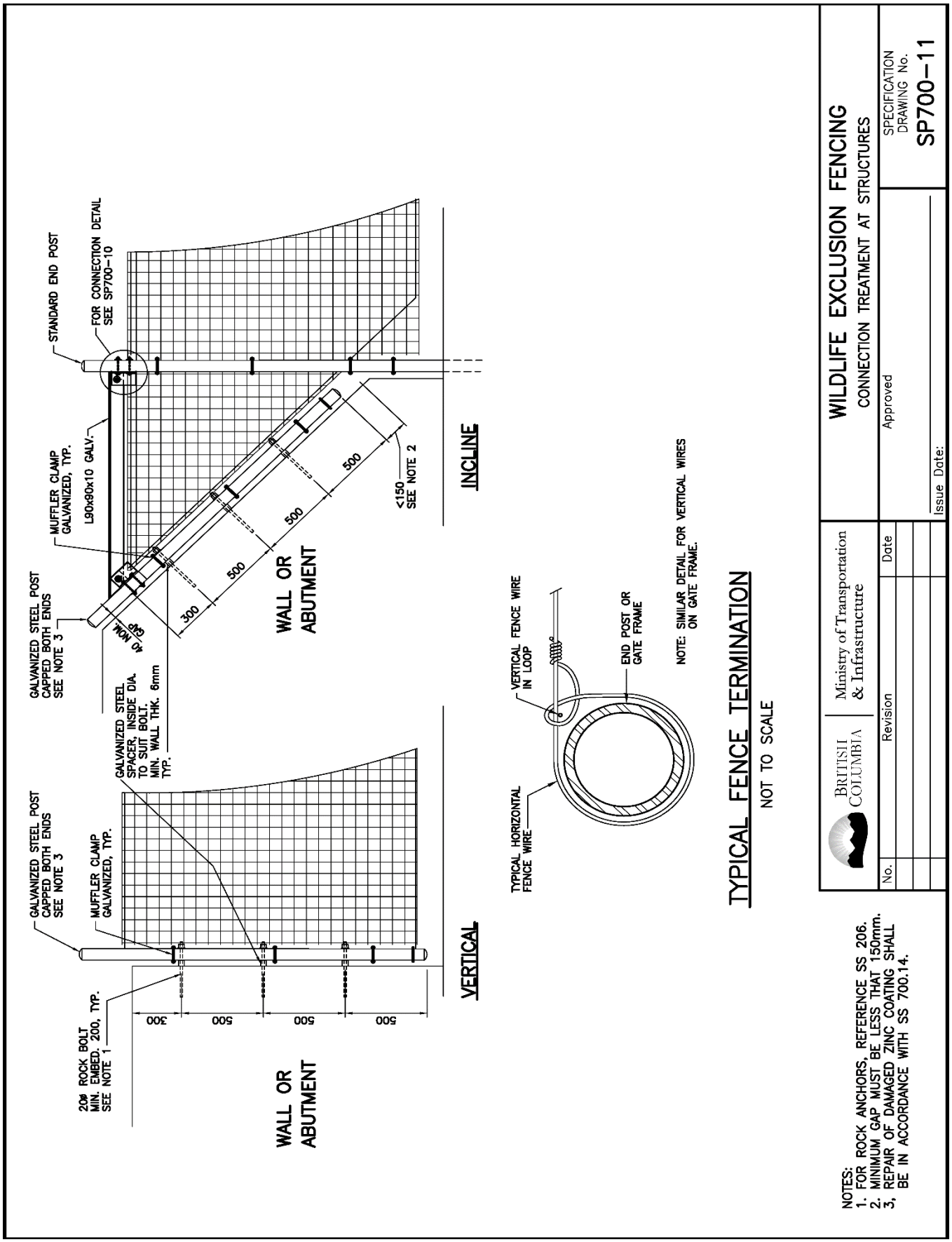
NOT TO SCALE

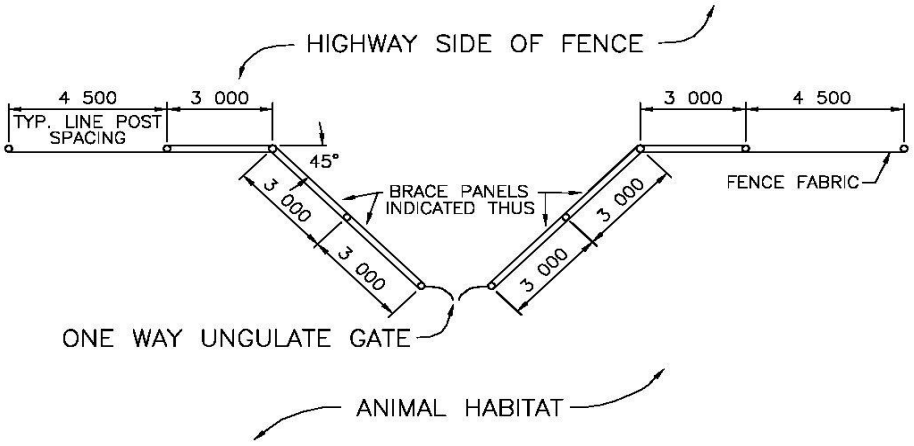
NOTES:
1. OVERSIZED HOLES SHALL BE COMPLETELY BACKFILLED WITH CONCRETE OR WELL TAMPED GRAVEL AROUND CONCRETE FOOTING.
2. CONCRETE MUST CONFORM TO SS 218, CLASS C (20mm COARSE AGGREGATE).

BRITISH COLUMBIA | Ministry of Transportation & Infrastructure	WILDLIFE EXCLUSION FENCING EMBEDMENT DETAILS FOR SOFT ORGANIC SOILD OR WETLANDS	
No. / Revision / Date	Approved Issue Date:	SPECIFICATION DRAWING No. SP700–08


LINE POSTS
BRACE POSTS
CORNER POSTS
END POSTS
25
450
100ø MIN. FOR NON SHRINK GROUT
200ø MIN. FOR CONCRETE
CONCRETE OR
NON-SHRINK GROUT
25
600
100ø MIN. FOR NON SHRINK GROUT
200ø MIN. FOR CONCRETE
CONCRETE OR
NON-SHRINK GROUT
GALVANIZED STEEL POSTS IN ROCK
NOT TO SCALE
NOTES:
1. CONCRETE MUST CONFORM TO SS 218, CLASS C (20mm COARSE AGGREGATE).
2. NON SHRINK GROUT SHALL BE A PRE BAGGED MIX WITH A MIN. COMPRESSIVE STRENGTH OF 20 MPa AT 28 DAYS.
BRITISH COLUMBIA
Ministry of Transportation & Infrastructure
No.
Revision
Date
WILDLIFE EXCLUSION FENCING
EMBEDMENT DETAILS FOR FENCING IN ROCK
Approved
Issue Date:
SPECIFICATION DRAWING No.
SP700-09


WIRE FABRIC FASTENING
73ø STEEL POST GALV.
GALVANIZED MUFFLER CLAMP TO BE USED AT TOP AND EVERY 0.6m TO THE BOTTOM OF FENCE FABRIC. TIE WITH 9 Ga. (3.75ø) GALV. WIRE BETWEEN CLAMPS AT 300 o.c. MINIMUM 4 CLAMPS
TOP WIRE OF FABRIC
CLAMP VERTICAL WIRE ALSO WHERE POSSIBLE TO DO SO
GALVANIZED MUFFLER CLAMP CLAMP TO BE USED AT THE BOTTOM
BOTTOM WIRE OF FABRIC
CLAMP VERTICAL WIRE ALSO WHERE POSSIBLE TO DO SO
FASTENING DETAILS FOR STEEL POSTS
(IN ELEVATION)
NOT TO SCALE
ANGLE FASTENING
ELEVATION
L90x90x10 GALV. TO FIT SNUG AGAINST POST
9.5ø BOLTS c/w 39ø WASHERS AND NUTS AS INDICATED (GALVANIZED)
73ø STEEL POST GALV.
SECTION
L90x90x10 GALV. 150 LONG
70
40
38
40
2-9 Ga. (3.75ø) GALV. WIRES.
NOTES:
1. FOR ROCK ANCHOR, REFERENCE SS 206.
2. MINIMUM GAP MUST BE LESS THAN 150mm
3. REPAIR OF DAMAGED ZINC COATING SHALL BE IN ACCORDANCE WITH SS 700.14.
BRITISH COLUMBIA
Ministry of Transportation & Infrastructure
No.
Revision
Date
WILDLIFE EXCLUSION FENCING
FASTENING DETAILS FOR STEEL POSTS
Approved
Issue Date:
SPECIFICATION DRAWING No.
SP700-10


STANDARD END POST
FOR CONNECTION DETAIL SEE SP700-10
MUFFLER CLAMP GALVANIZED, TYP.
L90x90x10 GALV.
GALVANIZED STEEL POST CAPPED BOTH ENDS SEE NOTE 3
40 NOM. GAP
GALVANIZED STEEL SPACER, INSIDE DIA. TO SUIT BOLT. MIN. WALL THK. 6mm TYP.
300
500
500
500
<150 SEE NOTE 2
WALL OR ABUTMENT
INCLINE
GALVANIZED STEEL POST CAPPED BOTH ENDS SEE NOTE 3
MUFFLER CLAMP GALVANIZED, TYP.
20ø ROCK BOLT MIN. EMBED. 200, TYP. SEE NOTE 1
300
500
500
500
WALL OR ABUTMENT
VERTICAL
VERTICAL FENCE WIRE IN LOOP
END POST OR GATE FRAME
TYPICAL HORIZONTAL FENCE WIRE
NOTE: SIMILAR DETAIL FOR VERTICAL WIRES ON GATE FRAME.
TYPICAL FENCE TERMINATION
NOT TO SCALE
NOTES:
1. FOR ROCK ANCHORS, REFERENCE SS 206.
2. MINIMUM GAP MUST BE LESS THAT 150mm.
3, REPAIR OF DAMAGED ZINC COATING SHALL BE IN ACCORDANCE WITH SS 700.14.
BRITISH COLUMBIA
Ministry of Transportation & Infrastructure
No. Revision Date
WILDLIFE EXCLUSION FENCING
CONNECTION TREATMENT AT STRUCTURES
Approved
Issue Date:
SPECIFICATION DRAWING No.
SP700-11



PLAN VIEW
SINGLE ONE WAY UNGULATE GATE LAYOUT
NOT TO SCALE

NOTES:

1) NYLON, OR OTHER SUITABLE PLASTIC, BUSHINGS SHALL BE INSTALLED ON THE GATE HINGES, TO LAY AGAINST THE MOUNTING PLATE.

2) 38mm X 50mm PATCHES OF 3mm NEOPRENE RUBBER SHALL BE PERMANENTLY ATTACHED TO EACH SWINGING SECTION, USING A SILICON ADHESIVE, SO THAT THE GATE CAN BE SLAMMED SHUT WITHOUT NOISE.

3) ALL METAL IN THE GATE ASSEMBLY SHALL BE GALVANIZED.

4) THE GATE TINES ONLY SHALL BE HEAT TREATED TO ROCKWELL HRC 30 TO 35.

5) ACID BATH TREATMENT OF THE TINES PRIOR TO GALVINIZING SHALL BE CONTROLLED TO MINIMIZE POTENTIAL WEAKENING OF WELDS.

6) GATES SHALL BE CAREFULLY FABRICATED AND WELDED TO AVOID BINDING. COMPLETED GATES SHALL BE TEST ASSEMBLED IN SHOP, ADJUSTED AND LUBRICATED TO SWING FREELY AND QUIETLY SHUT FROM ANY POSITION.

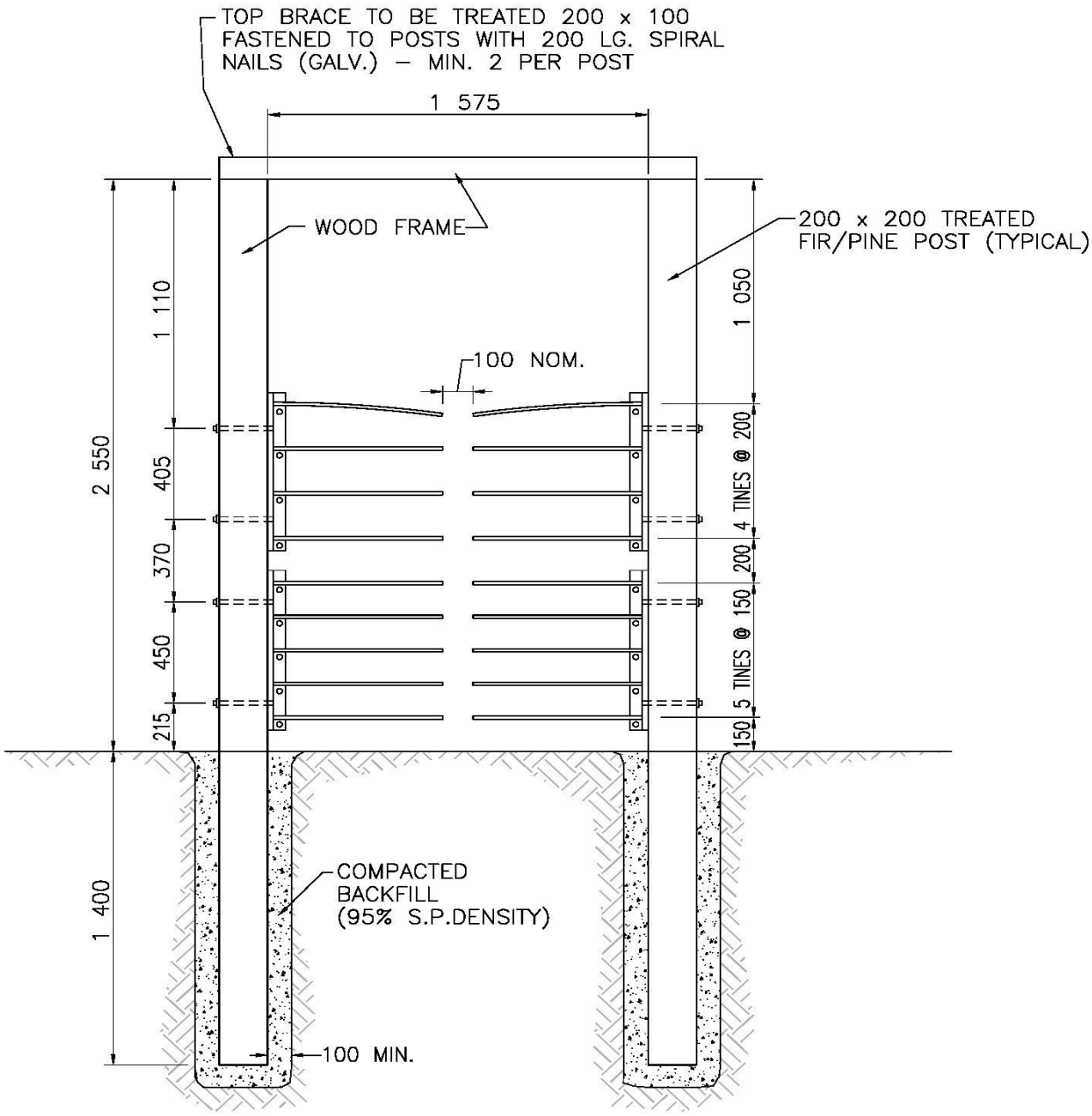
7) ALL STEELWORK TO BE GALVANIZED AFTER FABRICATION IN ACCORDANCE WITH SS422. GALVANIZING TO BE IN ACCORDANCE WITH ASTM A153 AND A123.

8) DRAWING TO BE READ IN CONJUNCTION WITH SP700–12.2 TO 12.7

9) BACKFILL TO BE a) IN FREE DRAINING NON–COHESIVE SOILS – NATIVE MATERIALS
b) IN COHESIVE SOILS – MIX NATIVE MATERIAL 1 TO 1 WITH COARSE ROCK AGGREGATE (MAX 75mm ø)

10) DRAWINGS TO BE READ IN CONJUNCTION WITH STANDARD SPECIFICATION SECTION 422 MISCELLANEOUS STEELWORK.

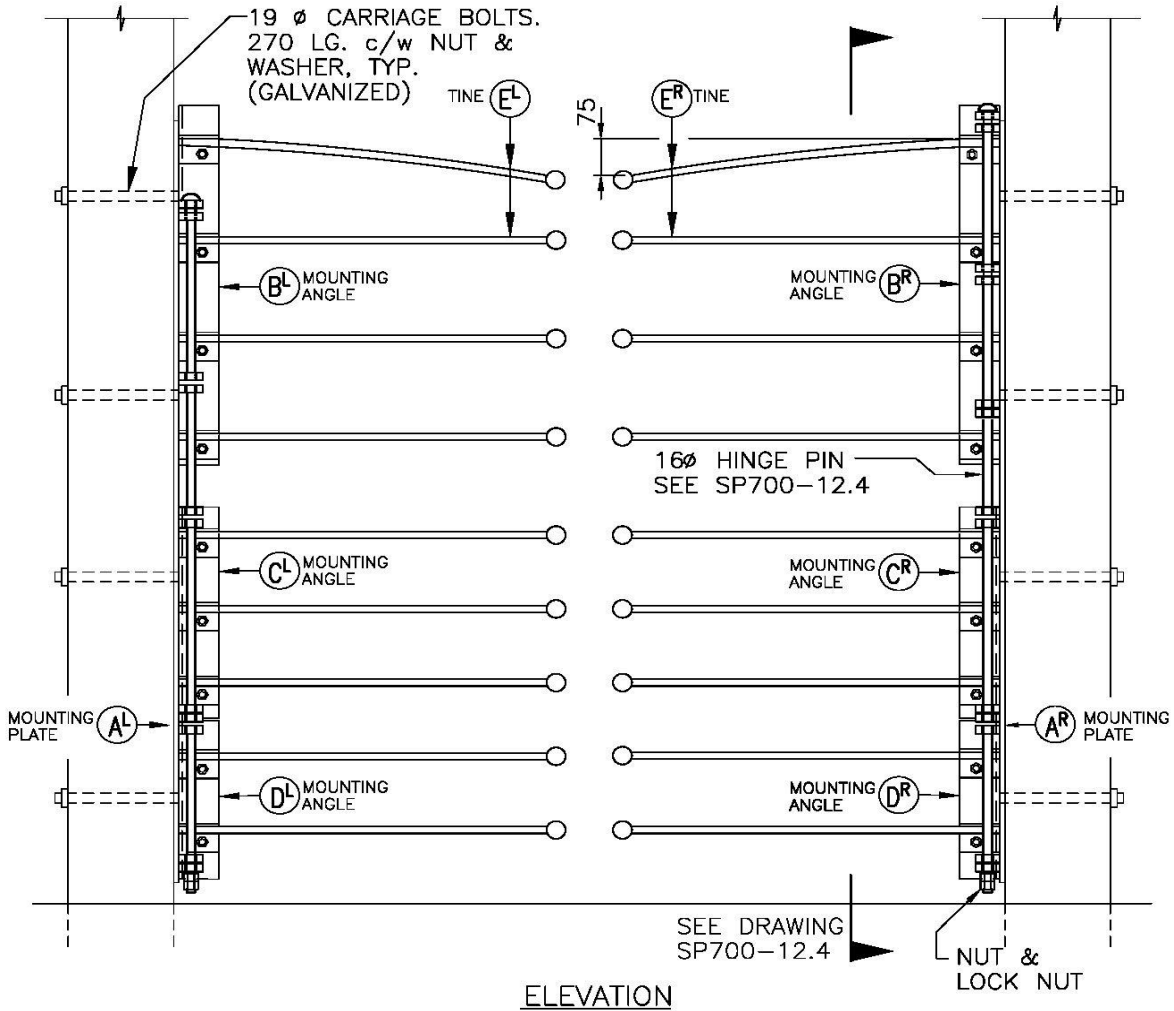
BRITISH COLUMBIA — Ministry of Transportation & Infrastructure	ONE WAY UNGULATE GATE UNGULATE GATE SHOP DRAWING ASSEMBLY LAYOUT	
No. / Revision / Date	Approved Issue Date:	SPECIFICATION DRAWING No. SP700–12.1



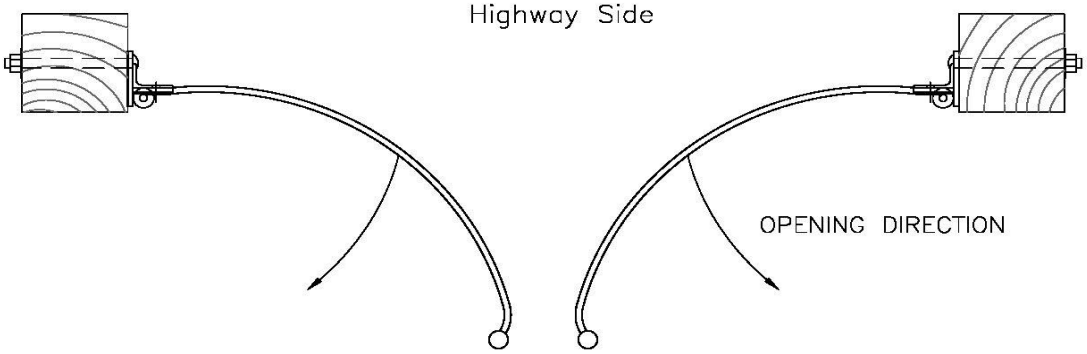
ONE WAY UNGULATE GATE ASSEMBLY

NOT TO SCALE

BRITISH COLUMBIA — Ministry of Transportation & Infrastructure	ONE WAY UNGULATE GATE UNGULATE GATE SHOP DRAWING ASSEMBLY LAYOUT	
No. / Revision / Date	Approved Issue Date:	SPECIFICATION DRAWING No. SP700–12.2



ELEVATION

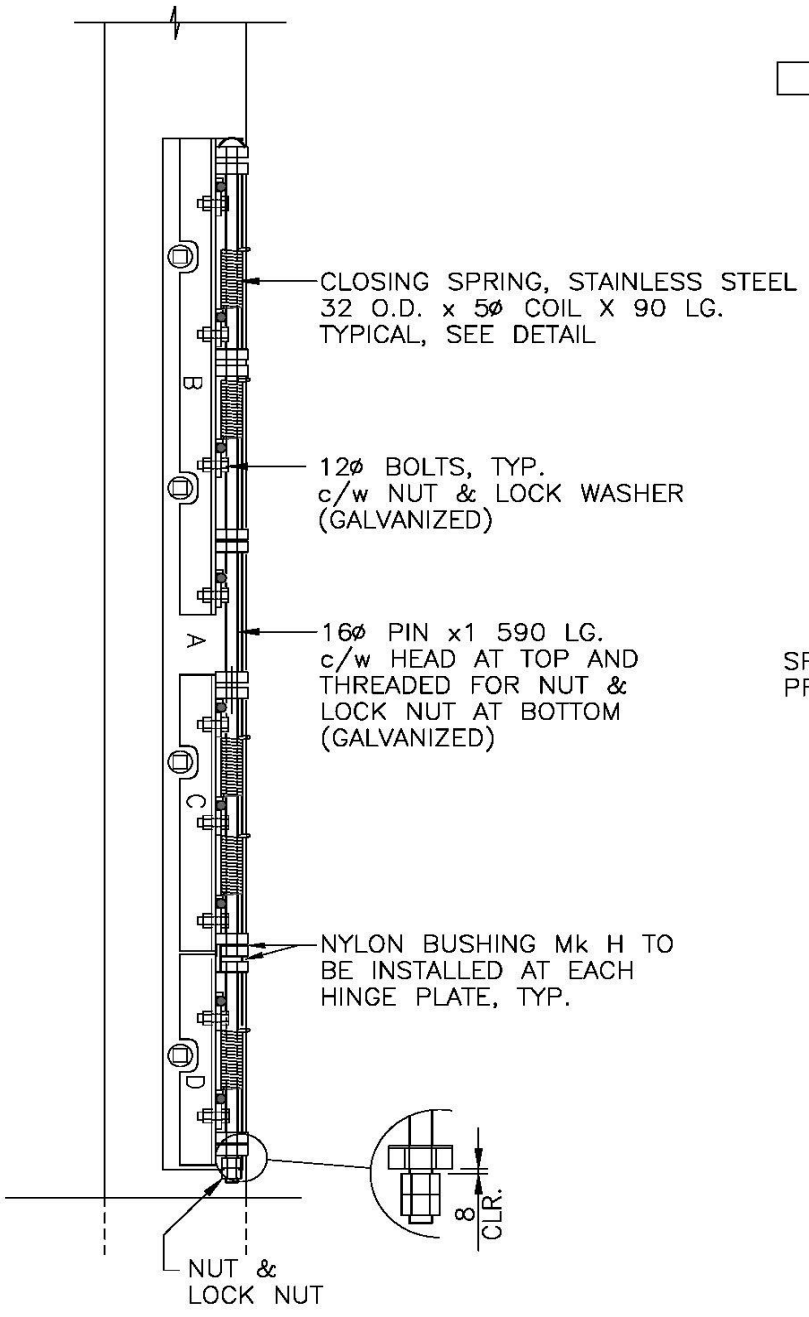


PLAN

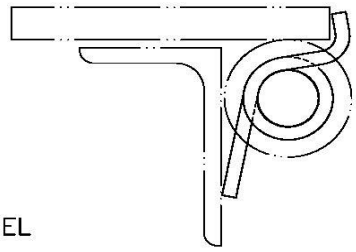
GENERAL ARRANGEMENT

NOT TO SCALE

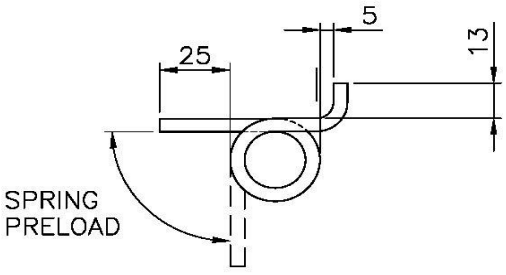
BRITISH COLUMBIA / Ministry of Transportation & Infrastructure			ONE WAY UNGULATE GATE GENERAL GATE ASSEMBLY
No.	Revision	Date	Approved
			Issue Date:
			SPECIFICATION DRAWING No. SP700–12.3



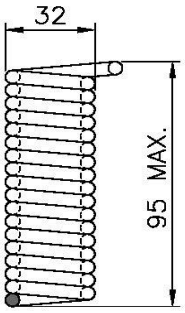
SECTION



IN PLACE PLAN



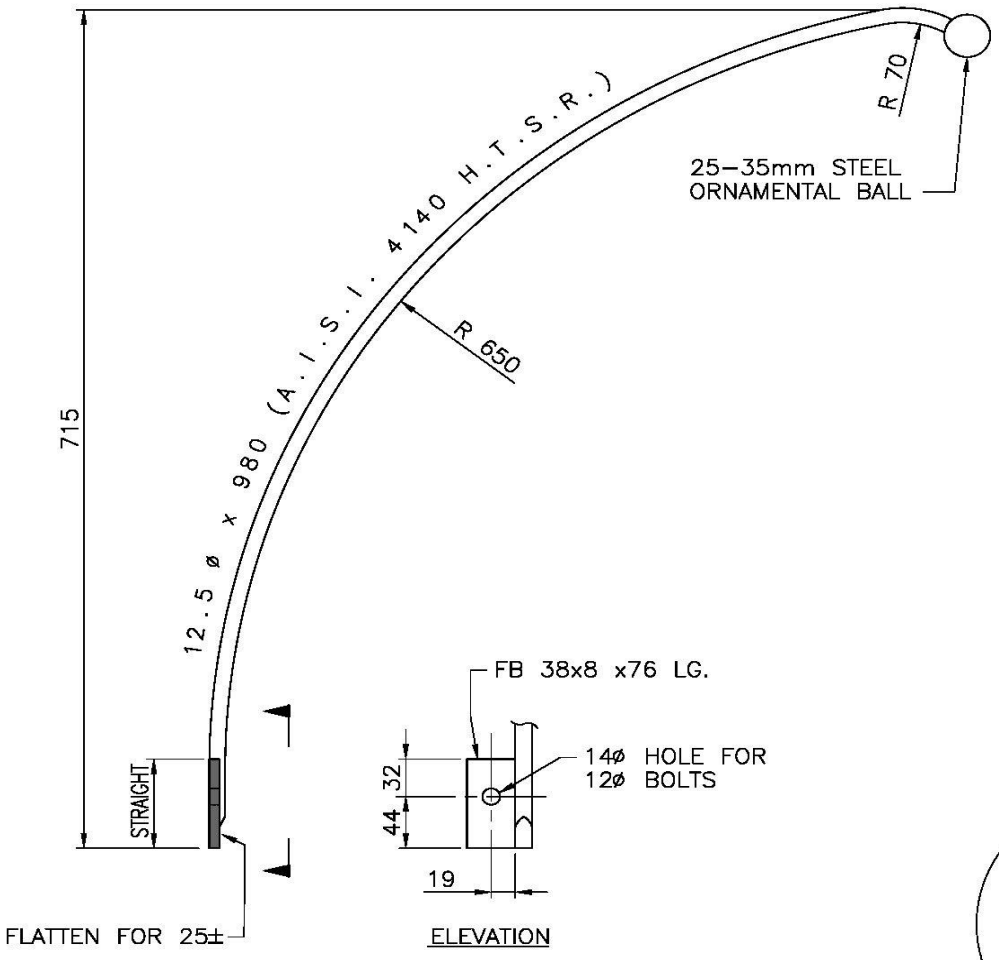
PLAN



ELEVATION

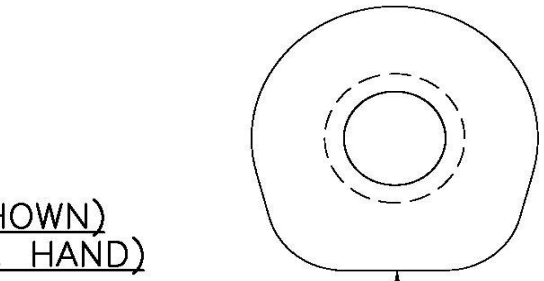
CLOSING SPRING
PROPOSED LAYOUT
10 REQUIRED

BRITISH COLUMBIA — Ministry of Transportation & Infrastructure	ONE WAY UNGULATE GATE GATE ASSEMBLY – SIDE VIEW	
No. / Revision / Date	Approved Issue Date:	SPECIFICATION DRAWING No. SP700–12.4

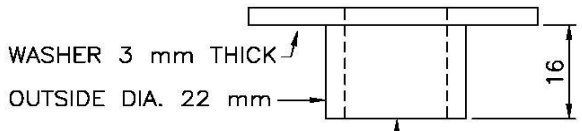


9 REQUIRED – TINE Mk E^R (SHOWN)
9 REQUIRED – TINE Mk E^L (OPP. HAND)

NOT TO SCALE



TOP VIEW



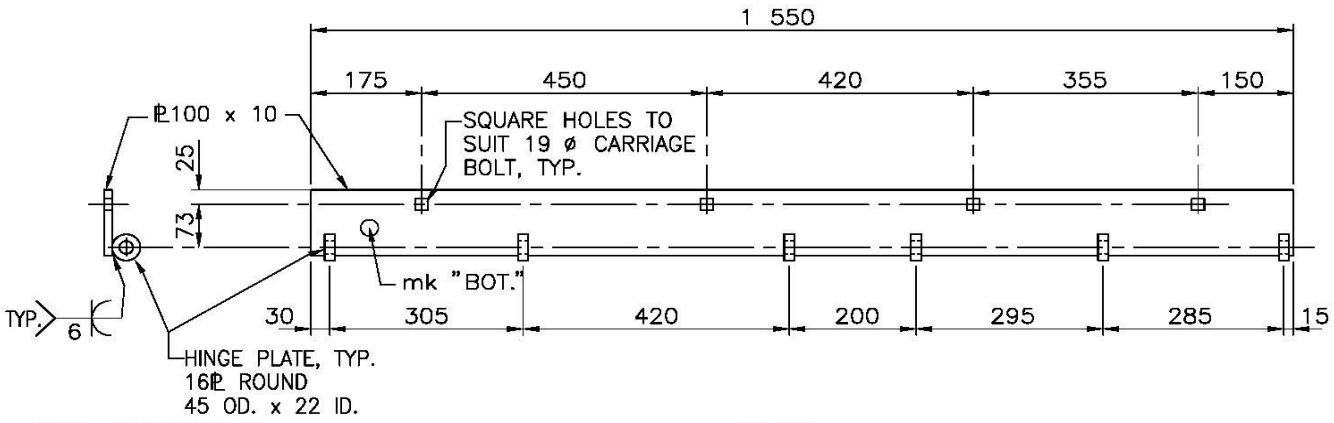
SIDE VIEW

NYLON BUSHINGS Mk H

NOT TO SCALE

22 REQUIRED

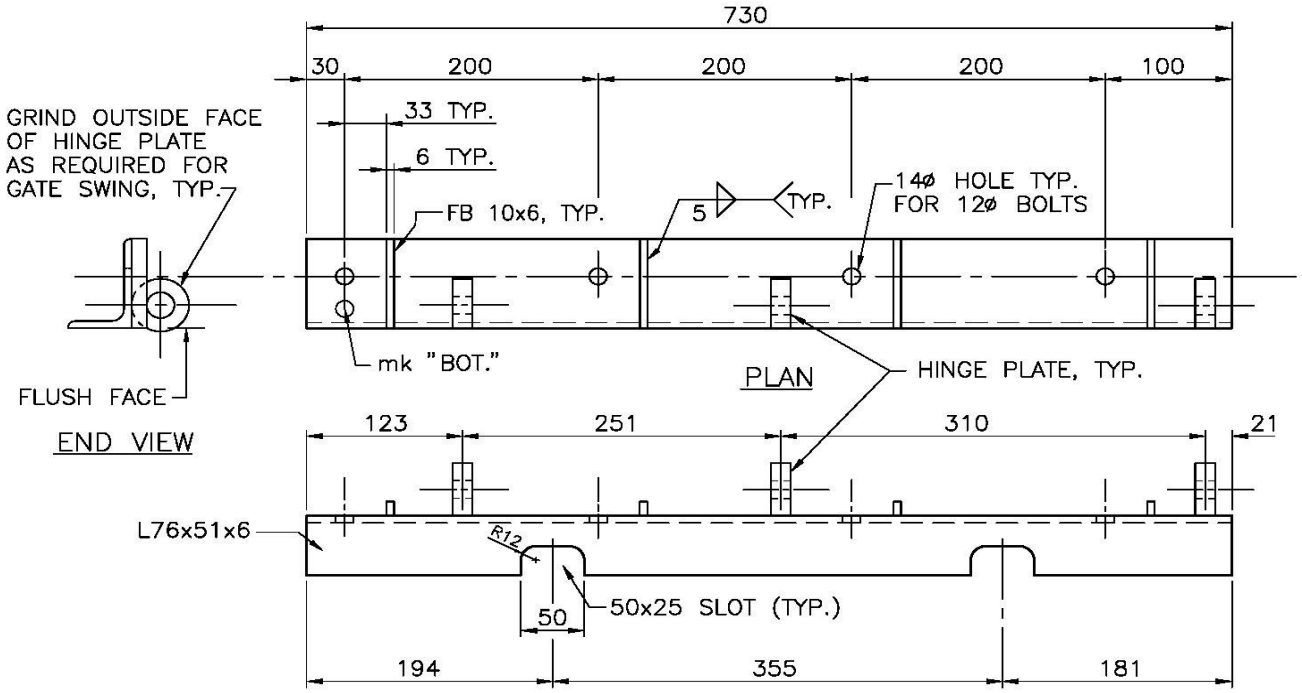
BRITISH COLUMBIA / Ministry of Transportation & Infrastructure	ONE WAY UNGULATE GATE TINE & BUSHING	
No. / Revision / Date	Approved Issue Date:	SPECIFICATION DRAWING No. SP700–12.5



END ELEVATION PLAN

ONE REQUIRED – MOUNTING PL – Mk. A1^{R} (SHOWN)
ONE REQUIRED – MOUNTING PL – Mk. A1^{L} (OPP. HAND)

NOT TO SCALE

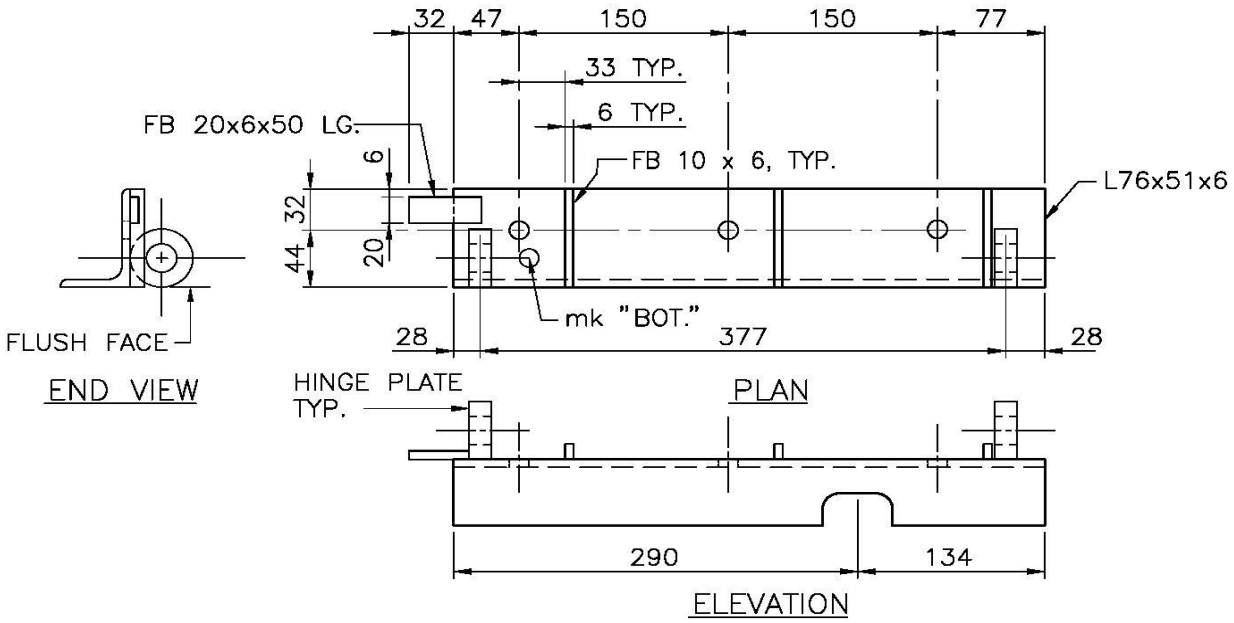


ELEVATION

ONE REQUIRED – ANGLE Mk B^{R} (SHOWN)
ONE REQUIRED – ANGLE Mk B^{L} (OPP. HAND)

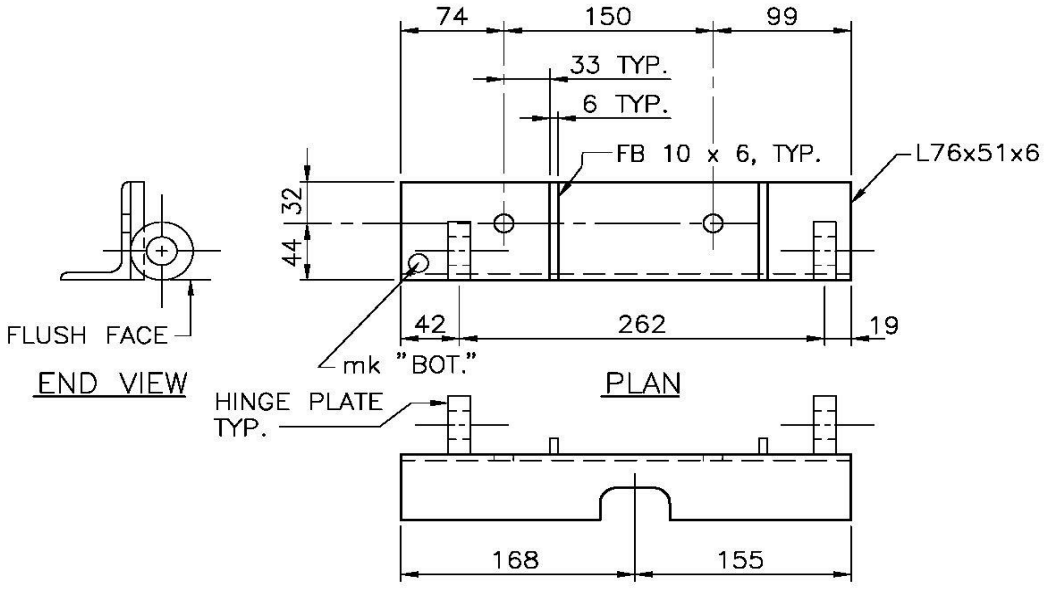
NOT TO SCALE

BRITISH COLUMBIA / Ministry of Transportation & Infrastructure			ONE WAY UNGULATE GATE MOUNTING BRACKETS Mk. A & B
No.	Revision	Date	Approved Issue Date:
			SPECIFICATION DRAWING No. SP700–12.6



ONE REQUIRED – ANGLE Mk. C^R (SHOWN)
ONE REQUIRED – ANGLE Mk. C^L (OPP. HAND)

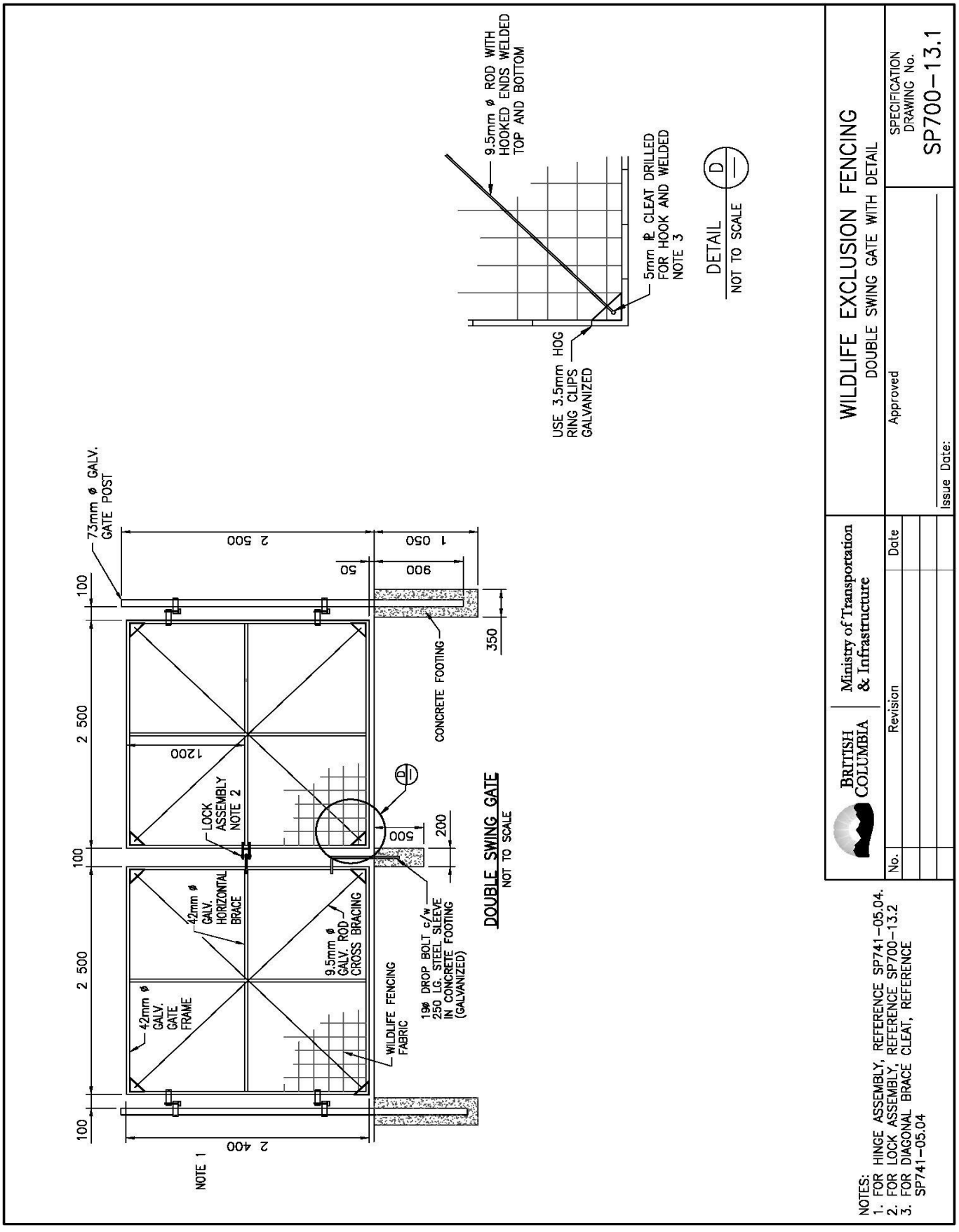
NOT TO SCALE

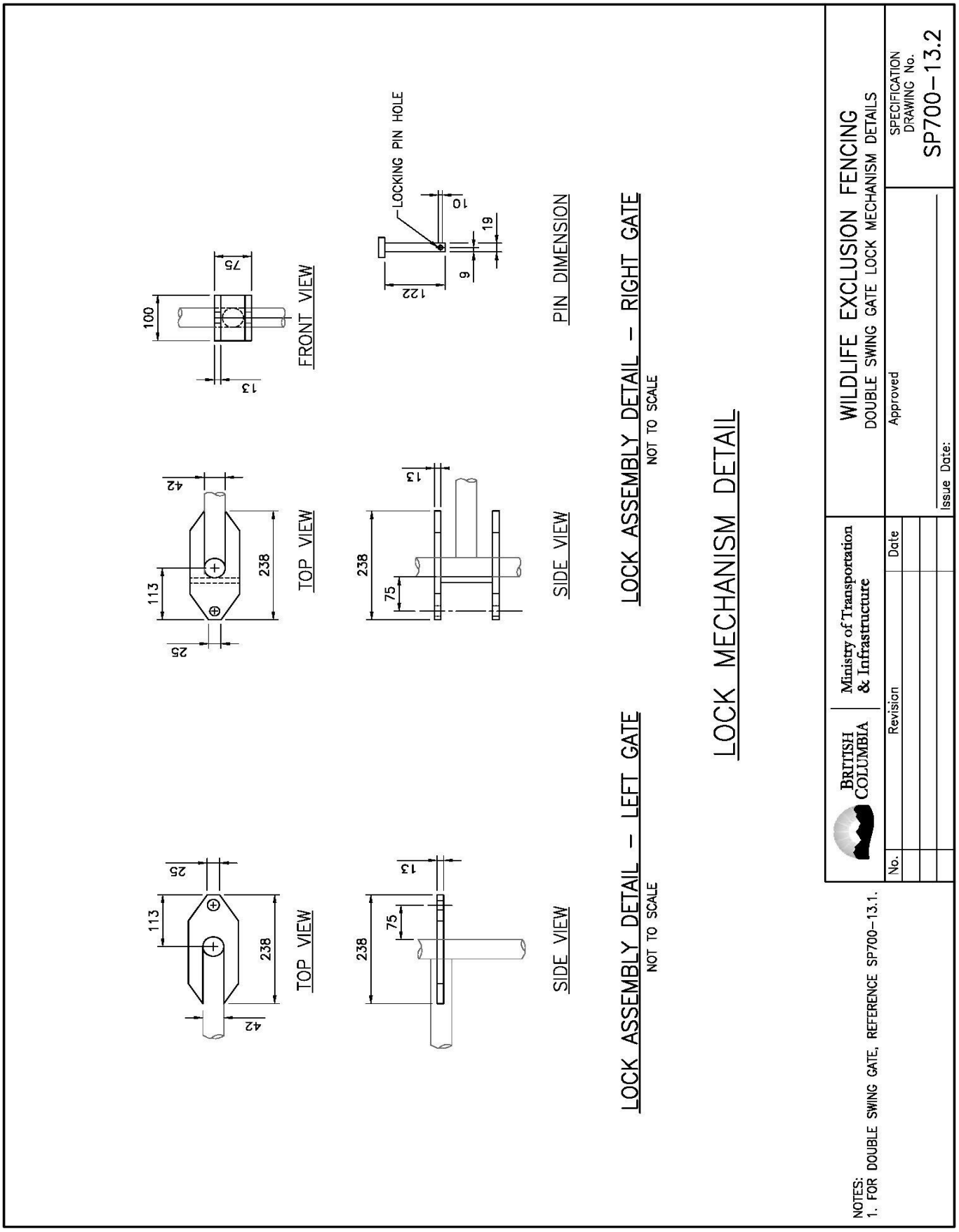


ONE REQUIRED – ANGLE Mk D^R (SHOWN)
ONE REQUIRED – ANGLE Mk D^L (OPP. HAND.)

NOT TO SCALE

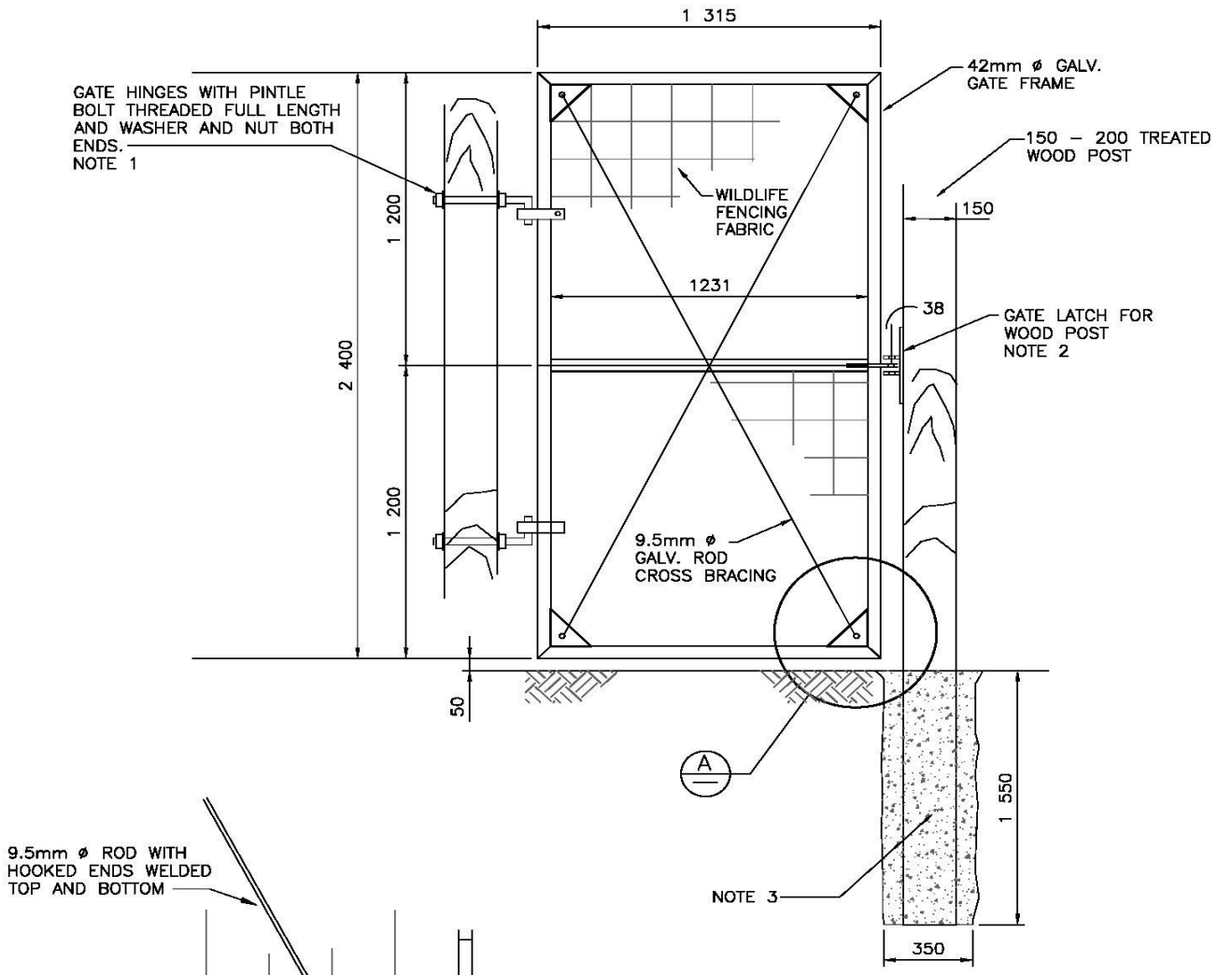
BRITISH COLUMBIA | Ministry of Transportation & Infrastructure			ONE WAY UNGULATE GATE MOUNTING BRACKETS Mk. C & D
No.	Revision	Date	Approved Issue Date:
			SPECIFICATION DRAWING No. SP700–12.7


9.5mm ø ROD WITH HOOKED ENDS WELDED TOP AND BOTTOM
5mm ℄ CLEAT DRILLED FOR HOOK AND WELDED NOTE 3
USE 3.5mm HOG RING CLIPS GALVANIZED
DETAIL D
NOT TO SCALE
73mm ø GALV. GATE POST
2 500
1 050
50
900
100
2 500
100
2 500
100
2 400
NOTE 1
1200
LOCK ASSEMBLY NOTE 2
42mm ø GALV. HORIZONTAL BRACE
42mm ø GALV. GATE FRAME
9.5mm ø GALV. ROD CROSS BRACING
WILDLIFE FENCING FABRIC
19ø DROP BOLT c/w 250 LG. STEEL SLEEVE IN CONCRETE FOOTING (GALVANIZED)
CONCRETE FOOTING
350
500
200
DOUBLE SWING GATE
NOT TO SCALE
NOTES:
1. FOR HINGE ASSEMBLY, REFERENCE SP741-05.04.
2. FOR LOCK ASSEMBLY, REFERENCE SP700-13.2
3. FOR DIAGONAL BRACE CLEAT, REFERENCE SP741-05.04
BRITISH COLUMBIA
Ministry of Transportation & Infrastructure
No.
Revision
Date
WILDLIFE EXCLUSION FENCING
DOUBLE SWING GATE WITH DETAIL
Approved
Issue Date:
SPECIFICATION DRAWING No.
SP700-13.1


LOCK MECHANISM DETAIL
LOCK ASSEMBLY DETAIL – LEFT GATE
NOT TO SCALE
TOP VIEW
SIDE VIEW
LOCK ASSEMBLY DETAIL – RIGHT GATE
NOT TO SCALE
FRONT VIEW
TOP VIEW
SIDE VIEW
PIN DIMENSION
LOCKING PIN HOLE
NOTES:
1. FOR DOUBLE SWING GATE, REFERENCE SP700–13.1.
BRITISH COLUMBIA
Ministry of Transportation & Infrastructure
No.
Revision
Date
WILDLIFE EXCLUSION FENCING
DOUBLE SWING GATE LOCK MECHANISM DETAILS
Approved
Issue Date:
SPECIFICATION DRAWING No.
SP700–13.2

LOCKABLE HUMAN ACCESS GATE

NOT TO SCALE



DETAIL A
NOT TO SCALE

NOTES:
1. GATE HINGE, REFERENCE SP741–04.04.
2. GATE LATCH FOR WOOD POST, REFERENCE SP700–14.2.
3. WELL TAMPED BACKFILLS (A) IN FREE DRAINING, NON–COHESIVE SOILS, USE NATIVE MATERIALS. (B) IN COHESIVE SOILS, MIX NATIVE MATERIALS 50–50 WITH COARSE ROCK AGGREGATE (MAX. 75 ø)

BRITISH COLUMBIA | Ministry of Transportation & Infrastructure

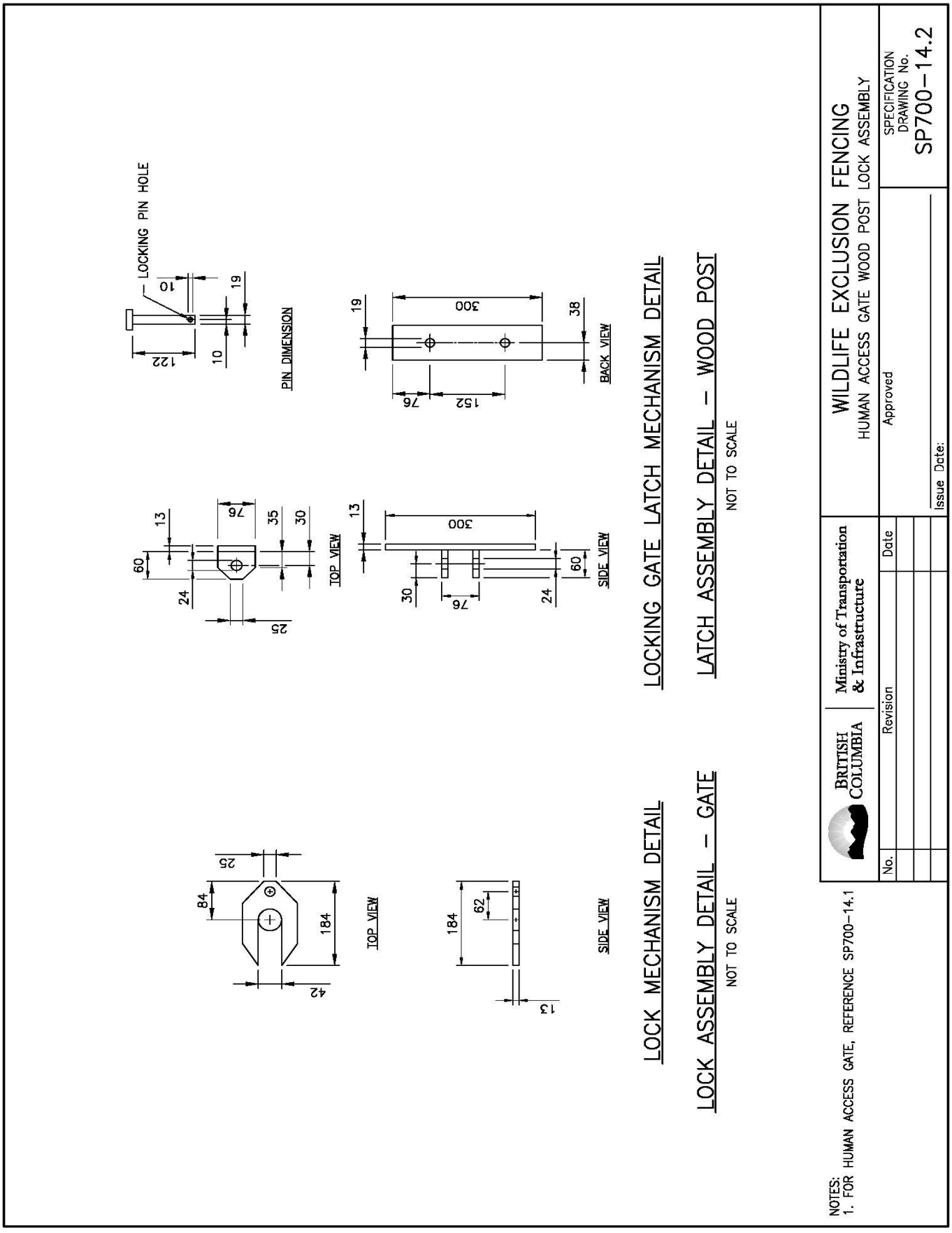
WILDLIFE EXCLUSION FENCING
HUMAN ACCESS GATE AND DETAIL

No.	Revision	Date

Approved

Issue Date:

SPECIFICATION DRAWING No.
SP700–14.1


LOCKING PIN HOLE
10
19
122
10
PIN DIMENSION
19
300
38
76
152
BACK VIEW
13
76
35
30
60
24
25
TOP VIEW
13
300
30
76
60
24
SIDE VIEW
LOCKING GATE LATCH MECHANISM DETAIL
LATCH ASSEMBLY DETAIL – WOOD POST
NOT TO SCALE
25
84
184
42
TOP VIEW
184
62
13
SIDE VIEW
LOCK MECHANISM DETAIL
LOCK ASSEMBLY DETAIL – GATE
NOT TO SCALE
NOTES:
1. FOR HUMAN ACCESS GATE, REFERENCE SP700–14.1
BRITISH COLUMBIA
Ministry of Transportation & Infrastructure
No.
Revision
Date
WILDLIFE EXCLUSION FENCING
HUMAN ACCESS GATE WOOD POST LOCK ASSEMBLY
Approved
Issue Date:
SPECIFICATION DRAWING No.
SP700–14.2

SECTION 741
FENCE CONSTRUCTION

DESCRIPTION

741.01 Scope – This Section covers the construction of wire fencing with wood and/or metal posts, gates, wood privacy and noise barrier fencing, and steel sidewalk fencing and, with reference to SS Drawings of the SP741 series, is intended to specify acceptable standards and some optional features as may be required by the Special Provisions.

The types of standard wire and chain link fencing covered by this Section are designated in Table 741-A.

Table 741-A: Types of Standard Wire and Chain Link Fencing

Type	Description	SP drawing
A	Special Wire Fabric Fence for use only on railway right-of-way	SP741-01.01
B	Standard Wire Fabric Fence	SP741-01.01
C	Standard Barbed Wire Fence	SP741-02.01
D	Chain Link Fence	SP741-05.01

Note: *While Type C fencing is the standard for highway right-of-way, the Ministry is working with an interagency group to develop a "wildlife-friendly" fence for use under appropriate conditions.*

See SS 700 for wildlife exclusion fencing.

MATERIALS

741.10 General – Steel and wire fence materials including wire fabric, barbed and high-tensile wire, chain link mesh and metal posts and rails are specified in SS 316.

Chain link fabric as protection on rock slopes is covered by SS 207 and SS 316.

All materials shall be supplied by the Contractor unless Ministry-supply is specified in the Special Provisions.

741.11 Treated Wood Materials

741.11.01 General – Wood posts, droppers, and braces shall be supplied by the Contractor in accordance with the requirements of SS 909.

Treated wood products used in fence construction shall meet the requirements of SS 908 as well as the product specific requirements in the following related Standard Specification sections:

- SS 316 Steel and Wire Fence Materials
- SS 700 Wildlife Exclusion Fencing
- SS 909 Treated Wood Fence Posts

741.11.02 Posts

(a) Round Wood Post Material – Treated Round Wood posts shall be prepared from straight peeled Lodgepole Pine or Jack Pine.

(b) Dimensional Tolerances – In addition to the requirements in SS 909, treated round wood posts shall have the following dimensions, each within a 2% tolerance:

(i) Length as necessary to comply with the applicable SS 741 SP Drawings, for the fence type specified and the soil type encountered;

(ii) minimum tip diameter of 150 mm; and

(iii) maximum butt diameter of 200 mm.

(c) Bottom Tapers – The bottoms of the round wood posts must be tapered to a rounded end for driveability.

741.11.03 Droppers – All droppers shall be free from knots and other imperfections injurious to strength.

Droppers shall be:

(a) 1100 mm ± 25 mm long for 4-wire Type C fences; and

(b) 1200 mm ± 25 mm for Type C and C2 fences.

Approved proprietary grooved wood droppers with necessary wire clips may be specified or approved by the Ministry Representative for high-tensile smooth-wire fencing.

Note: For Type C fencing, proprietary prefabricated galvanized sheet metal and clip droppers may be approved by the Ministry Representative as an alternative to the above wood droppers. Galvanized twisted wire fence stays may be specified or permitted only for the extension of normal fence heights to deer height and the like.

741.12 Standard Wire Type A, B & C Fences and Gates – Materials generally shall be in accordance with the requirements set out on SS Drawings SP741-01.01 and SP741-02.01, the relevant subsections of SS 316 and and/or the Special Provisions.

741.12.01 Gates – Gates shown on the Drawings shall be of the prefabricated type indicated on SS Drawings SP741-04.04 and SP741-04.05, and as specified in SS 316.11, and/or as specified in the Special Provisions.

741.13 High-tensile Smooth-wire Fences – Materials to be supplied include:

(a) high-tensile smooth galvanized wire 2.5 mm nominal diameter

(b) mechanical wire splices and fasteners for high-tensile wire

(c) in-line high-tensile wire tensioning devices

(d) metal or wood droppers for high-tensile fencing.

together with steel posts, gates and hardware, brace wire, dowels, staples and the like in accordance with the relevant subsections of SS 316.

741.14 Chain Link Type D Fences and Gates – Chain link mesh, steel pipe and accessory materials generally shall be in accordance with the requirements set out on SS Drawings SP741-05.01 and SP741-05.02, the relevant subsections of SS 316, and/or the Special Provisions.

741.14.01 Gates – Gates shown on the Drawings shall be of the type indicated on SS Drawings SP741-05.03 and SP741-05.04 and/or specified by the Special Provisions, and shall be equipped with hinges (or sliding) and locking hardware.

741.15 Wood Fences – Materials used for the fabrication and installation of wood fences shall be as follows:

741.15.01 Concrete Footings – All concrete used as post anchorage backfill shall be well-consolidated Class C concrete conforming to SS 218 Table 218-A, or a non-shrink grout, minimum compressive strength of 20 MPa at 28 days, approved by the Ministry Representative**.**

741.15.02 Lumber – Rough sawn or dressed, as specified, shall be of Western Red Cedar with:

(a) Wood posts and stringers of "No. 1 Structural" grade posts and framing or plank

(b) Boards and planks of "Quality Fencing" grade except where noise barrier fencing is specified for "Select Fencing" or plank equivalent.

741.15.03 Galvanized steel pipe – 48 mm OD for privacy fence posts shall conform to the requirements of SS 316.10 with weatherproof caps where open ends are not covered by wood members.

741.15.04 Steel – Steel used as posts for noise barrier fences or as wood post supports shall be of the structural shape(s) indicated in accordance with CSA G40.21, Grade 300W and where galvanized, hot dipped to the requirements of ASTM A123.

741.15.05 Fastenings: Bolts generally shall conform to ASTM A307, nuts to ASTM A563 Grade A, plain washers to Type A, plate washers, where required, to ASTM A36; all galvanized according to CSA G164. Nails shall be casing headed heavy gauge of appropriate length, hot dip galvanized with deformed shank (annular, barbed or helical) for noise barrier fence plank fixing.

741.15.06 Finish – If required, penetrating stain with preservative shall be of type and colour specified in the Contract, and shall be applied to all surfaces after cutting to size, prior to prefabrication or installation, and on any cuts before final fitment.

CONSTRUCTION

741.30 Cutting and Boring Treated Wood Products – No cutting of pressure treated wood posts will be permitted without authorization of the Ministry Representative.

741.30.01 Cut at Top – When cutting is authorized, the cut must be only at the top of the round wood post.

741.30.02 Treatment – All cuts, borings and superficial damage shall be treated immediately with a preservative in accordance with SS 908.

741.31 Provision of Fencing – Fencing of the type(s) called for shall be carried out at the locations and as shown on the Drawings with the materials to the height, spacing and with accessories all in accordance with the details indicated on the Drawings, Standard Specifications, Special Provisions and SP Drawings or to the direction of the Ministry Representative.

All material shall be supplied by the Contractor except where supply in whole or in part by the Ministry DDP (Incoterms© 2020: Delivery Duty Paid) the Contractor's job site yard or Ministry's yard is specified.

Construction shall be carried out with all labour, tools, equipment and incidentals supplied by the Contractor, as necessary, to complete all fencing work in accordance with good work practice.

741.32 Clearing & Grading – All trees (other than any required by the Ministry Representative to remain), all brush and other obstacles which interfere with the construction and maintenance of fencing and not removed by the normal clearing operations, shall be removed prior to commencing fencing work so that both sides of the fence line are free of all clearing and grubbing debris.

The cleared and graded area must be a minimum of 3 m wide on each side of the fence to permit access for fence repairs and maintenance, unless a lesser width is permitted by the Ministry Representative. The graded area shall have a 2-5% cross-fall towards the open side of the slope to provide drainage. No fill shall be placed in any drainage path that crosses the fence line.

The ground line for the fence should be smooth and continuous for a minimum of 1 m on both sides of the fence. Minor ground undulations shall be corrected to obtain a smooth uniform grade, but appreciable grade depressions may be backfilled only with the permission of the Ministry Representative.

The site shall be left in a smooth and tidy condition.

741.33 Setting Out and Connections to Existing – Fence line, as shown on the Drawings, generally will be along the right-of-way boundary. Where undergrowth clearing is carried out to the right-of-way boundary, the fence line will be offset 0.5 m in from the boundary.

Post installation in fill material or minimum overburden shall be carried out to the Ministry Representative's directions.

Where it is not possible to drive or set wood posts to proper depth or to relocate same along the fence line, steel fence posts as specified or, where permitted, multiple wood post and brace assemblies shall be substituted.

Existing cross fences shall be connected to new with posts and braces for tensioning fencing wire in every direction in accordance with SS Drawings SP741-01.03 and SP741-02.02.

741.34 Standard Wire Type A, B & C Fences and Gates

741.34.01 Post Installation – Fence posts shall be driven in place with equipment acceptable to the Ministry Representative, set in augered pilot holes or, where permitted by the Ministry Representative, set in dug holes with necessary well tamped backfill for a firm installation and post penetration to at least the depth indicated on SS Drawings SP741-01.01 and SP741-02.01.

On straight alignments all posts shall be plumb. On appreciable grades, posts shall be installed perpendicular to the slope.

On curved alignments, the posts shall be set 50 mm off plumb away from the curve centre, with a post spacing in accordance with SS Drawing SP741-04.02 and increased post lengths and bury depth for post stability where necessary.

Gate post sizes and stabilizing shall be as required by the Special Provisions and/or SS Drawing SP741-04.04.

Steel fence posts, as specified by SS 316.09, are required on exposed rock or rock with "minimum overburden" (as defined on SS Drawing SP741-04.01) and shall be driven and/or wedged to the full depth in a vertical drilled hole of minimum diameter or set plumb and rigid in cement and sand or fine aggregate mortar all as indicated by SS Drawing SP741-04.01.

Note: Assemblies of securely wire tied multiple wood posts/braces may be specified or permitted by the Ministry Representative especially for minor rock outcropping and unforeseen minimum overburden occurrences.

When full bury depth of wood posts is not attainable, the specified steel fence posts shall be used to the full penetration into rock, all as indicated on SS Drawing SP741-04.01.

Tops of all posts shall be set to a uniform 50 mm above the level of the top wire of fencing and, where necessary, cut to line-up in vertical uniformity after inspection by the Ministry Representative. Wood post tops where cut, shall be bevelled and preservative treated in two heavy applications.

Wood posts which are burred, split or otherwise damaged from the installation, and which are not acceptable to the Ministry Representative shall be replaced at the Contractor's expense.

741.34.02 Bracing for Fence Tensioning – Braces of treated wood and twisted galvanized diagonal wire shall be installed between end posts and adjoining panel posts to stabilize the tensioning of the fencing fabric and/or barbed wire, all as indicated by SS Drawings SP741-01.03 and SP741-02.03. Corner and intersection assemblies shall be similarly tensioned in each fence direction.

Intermediate tensioning assemblies shall be provided within the maximum spacings indicated on SS Drawings SP741-01.02 and SP741-02.02 to make use of rolls of fabric etc. of standard length with a minimum of cutting and waste.

Steel corner, intersection and intermediate tensioning assemblies with steel angle posts and diagonal bracing are indicated on SS Drawing SP741-04.01.

Horizontal alignment changes over 30° with wood posts and over 15° with steel posts shall be stabilized as for corners above. Where a change is less than 30° with wood posts, a pair of line posts, set 2.4 m or 3.0 m apart, shall be stabilized with a horizontal wood brace and diagonal wire bracing both ways. For a change of less than 15° with steel posts, a steel line post at the change shall be stabilized in both directions with diagonal steel braces to adjoining line posts as indicated on SS Drawing SP741-04.01.

Similar requirements shall be maintained at each change to or within any curved fence alignment where the closer post spacing indicated by SS Drawing SP741-04.02 is not considered adequate.

Note: The previous requirements for alignment changes may be waived by the Ministry Representative wherever, from consideration of the soil conditions, the Ministry Representative directs that the stability and strength of the fence is not likely to be appreciably lessened by such change of alignment.

Vertical alignment changes shall be stabilized, and fence requirements at ditch and gully conditions shall be as called for by SS Drawing SP741-04.03 or as the Ministry Representative may direct.

741.34.03 Fabric and Wire – Fence fabric shall normally be installed on the side of the posts away from the highway, stretched between end type assemblies and intermediate tensioning assemblies with proper equipment (tensioning directly by truck or tractor will not normally be permitted) and securely stapled in accordance with good practice to wood posts, as indicated by SS Drawing SP741-01.01, or securely wired or clipped to steel posts to permit free wire movement on line posts. Similarly, barbed wire shall be installed so as to allow it to "prestretch" before final tensioning, and shall be securely stapled to tensioning assemblies.

For fencing on a curve requiring the fabric and/or barbed wire on the highway side of the posts, all line wires shall be double stapled, and the top wire shall be securely wire tied to posts in addition where the adjoining property has livestock.

Barbed wire Type C fences shall normally have wood droppers "interwoven" and securely figure-of-eight wire tied to every line wire, as indicated by SS Drawing SP741-02.01. Alternatively, proprietary galvanized sheet metal

prefabricated droppers of pre-approved design for secure clip-on application may be approved.

Note: High livestock pressures may warrant Type C1 fencing and nursery livestock enclosures Type C1 or B fencing. Where such fences, 1200 mm and higher, are in wildlife sensitive areas, the top strand may be specified smooth (barbless) as Type C2 or B1 fences.

741.34.04 Gates – Hardware, of design to permit the gate to operate correctly, shall be securely attached to prevent the easy removal of the gate and hardware. Hinges shall be installed to permit the gate to swing back one-way against the fence. Locking hardware shall be of the type specified.

741.35 741.35 High-tensile Smooth-wire Fences – For rangeland and other locations with favourable terrain, soil conditions and fence alignment at the locations indicated on Drawings, construct high-tensile smooth-wire fencing between rigid end, gates, corner and any necessary intermediate tensioning assemblies in general accordance with SS Drawings SP741-03.01 through SP741-03.03.

741.35.01 Fence Style – Fence style for wildlife crossing areas shall comprise 5 wires, the bottom wire for installation at 250 mm ± 50 mm above grade at and between posts, the next two wires at 200 mm spacing and the top two 225 mm for a total height of 1100 mm, as indicated by SS Drawing SP741-03.01 for HT Fences. **Note:** This fence style is under review by a multi-agency group, developing "wildlife-friendly" configuration(s). Check with the Ministry's Environmental Management Unit for the appropriate configuration.

Elsewhere, and for nursery livestock enclosures and other areas of high livestock pressure, a 6-wire fence may be specified, having the bottom wire 200 mm ± 50 mm above grade at and between posts with the remaining wires at 200 mm spacing for a total height of 1200 mm indicated as HT-1 Fences.

741.35.02 Post and Brace Installation – Fence posts shall be driven in place with equipment acceptable to the Ministry Representative and set out in general accordance with the recommendations of the Ministry of Agriculture "Fencing Factsheet 307.50-1 (Dec. 2015)", good local trade practice and to the Ministry Representative's direction.

Terminal (end and gate) tensioning assemblies and any intermediate assemblies shall be "single" assemblies and at the spacings and post bury depths indicated by SS Drawing SP741-03.02 (or such greater depths and/or decreased spacings as may be directed by Ministry Representative) with "double" assemblies used with scant bury depths all to form rigid, stable and accurately aligned assemblies for tensioning line wires.

Line posts 100 mm minimum diameter driven to a depth of not less than 650 mm in firm soil, or as otherwise required by SS Drawing SP741-03.01, or directed, shall be spaced up to a maximum of 15 m where permitted, but normally at 10 m spacing and such closer spacing at changes in horizontal and vertical alignments to keep the bottom wire at the regular specified ground clearance.

741.35.03 Wire and Dropper Installation – Wires shall be installed on the side of the fence posts away from the highway except on such curves requiring the wires on the highway side so as to bear on the posts.

Stapling doubled at curves, etc. and mechanical fasteners at tied-off ends and splices (where permitted) shall be to the Ministry of Agriculture's recommended practice.

With supplied tensioning devices installed in each line, all wires shall be individually tensioned in stages and after "wire-set" to 1.33 kN (300 lbs) at completion.

Droppers, as supplied, shall be installed and securely clipped to all line wires at 3 to 4 m spacing.

741.36 Chain Link Type D Fences and Gates

741.36.01 Post and Rail Installation

(a) All terminal posts (posts at ends, gates, corners and intersections), all line posts and any intermediate tensioning posts shall be set plumb into concrete footings in augered or dug holes to the depths and regular spacing all as shown on SS Drawing SP741-05.01, or as otherwise specified or directed.

(b) Gate post sizes and stabilizing shall be as required by the Special Provisions and/or SS Drawing SP741-05.03.

(c) On exposed rock, posts shall be set without concrete footings to full depth and fully grouted in holes, 25 mm greater than the pipe diameter, drilled to a depth into solid rock of half the depth of pipe bury to that indicated for concrete footings on Drawing SP741-05.01 (i.e. c/2 and f/2)

Grout shall be non-shrink cement and sand mortar. Standard length posts may be used where the overburden depth is such that the depth of bury into solid rock is less than that specified above, provided normal diameter concrete footings of the diameter indicated on SS Drawing SP741-05.01 are formed from solid rock to grade level. Similarly, form concrete footings for posts set in loose or friable rock. Sleeves shall be provided to form holes similar in diameter and depth to above for casting into concrete where shown or specified.

(d) Tops of all posts shall be set or cut for an even height of top rail, which shall form a continuous brace and mesh support between terminal posts and any intermediate tensioning posts. Top rails shall pass through line post caps and be joined in the length with internal sleeves to allow expansion and contraction. All posts shall be fitted with appropriate weathertight caps securely fixed.

741.36.02 Bracing for Fence Tensioning – Bottom tension wire shall be securely fixed taut and sag free to terminal posts and any intermediate tensioning posts. Similarly, provide top

tension wire, when specified, in place of top rail to pass through line post tops except on uneven terrain where the height of the top tension wire shall be such that it is secured to all line posts and the mesh within the top 300 mm of the mesh.

Terminal posts, where more than 10 m apart in any fence run, shall have horizontal pipe braces to adjoining line posts. Diagonal pipe braces may be called for where soil conditions warrant, especially where fencing without top rails is specified, but no pipe bracing is normally required for residential height fences.

Intermediate tensioning assemblies shall be provided where terminal posts are more than 150 m apart, and at any subsequent 150 m maximum spacing, to consist of a straining post (to full height of fence where barbed wire on extension arms is specified) with horizontal pipe braces at the mid to two-thirds height above grade to adjoining line posts each way for the discontinuity of top rail, tension and/or barbed wire and mesh; provide similar tensioning assembly at abrupt vertical alignment changes.

Horizontal alignment changes where abrupt shall be considered as corners.

At changes in horizontal alignment and to curved fence alignments, where the top rail can be continuous by accurately bending to proper curvature without damage to the galvanized coating, a pair of line posts shall be stabilized and tensioned as follows:

(a) A horizontal pipe brace shall be securely fixed to adjoining line posts at the two-thirds height above grade.

(b) Crossed diagonal wire braces shall each be two strands of 3.5 mm nominal diameter galvanized wire securely fixed to the brace band fixing of the horizontal brace and to a brace band 100 mm above grade.

(c) Each pair of wires shall be twisted taut to mutually stabilize the assembly with the tensioning battens left in place.

Similar requirements shall be maintained within any curved fence alignment over 30 m in length.

Note: The previous requirements may be waived by the Ministry Representative wherever the Ministry Representative directs that, from consideration of the soil conditions and footing sizes, especially where with continuous top rail, the stability and strength of the fence is not likely to be appreciably lessened by the change in alignment.

Fencing where continuous over a creek or ditch shall be stabilized with a horizontal pipe brace and diagonal brace wires as specified above, and the bottom tension wire shall be wire tied to a similar pipe brace. In addition, in-fill under where specified or required, shall be a hanging or rigidly fixed bent pipe of brace diameter hung or in-filled with closely spaced barbed wire or suitable fence fabric (similar to that indicated on SS Drawing SP741-04.03), all to the Ministry Representative's direction.

741.36.03 Fabric and Wire – Chain link fencing mesh shall be stretched between terminal posts and any intermediate tensioning posts with proper equipment (tensioning directly by truck or tractor will not normally be permitted), and secured with tension bars and bands, tie wire and clips all in accordance with the requirements shown on Drawing SP741.05.02. The mesh shall be installed on the highway side of the fence posts, or as otherwise specified or directed, and joints in the length shall be made by weaving the meshes together with a single wire picket to form a neat continuous fabric mesh.

Security barbed wire, where specified, shall be installed in the slots of all extension arms and secured to extended height terminal and any intermediate tensioning posts taut and free of sag.

741.36.04 Gates – Hardware, of design to permit the gate to operate correctly, shall be securely attached to prevent the easy removal of the gate and hardware. Hinged gates shall be installed to permit the gate to swing back one-way against the fence.

741.37 Wood Fences

741.37.01 Privacy Fences – Galvanized pipe posts shall be set plumb in concrete footings in augered or dug holes to the depth and regular spacing all as shown on SS Drawing SP741-06.02.

Wood fixing members for fence panels shall be securely nailed, "II-shaped" frames fixed and supported with 9.5 mm (3/8") galvanized bolts and nuts to pipe posts all as indicated on SS Drawing SP741-06.01.

Metal fixing members for fence panels, when specified, shall be of named or otherwise pre-approved proprietary brand or detailed formed metal channel and strap, bolted on around pipe posts and securely machine screwed and tapped at the correct levels all as indicated on SS Drawing SP741-06.03.

Fence panels of boards and stringers prestained, if specified, shall be prefabricated to the required design, as indicated by the Contract Drawings, Special Provisions and/or SS Drawings SP741-06.01 through SP741-06.03, and shall be accurately fitted and securely nailed to fixing members on steel posts to form uniformly level and/or stepped fencing.

741.37.02 Noise Barrier Fences – Posts of indicated type, size and length shall be set plumb in concrete footings in augered or dug holes to the depth and regular spacing all as shown on the Contract Drawings.

Stringers, to bear full width of post at each end of "alternating" or "zig zag" (on plan) panels, shall be securely bolted with large washers behind head and nut.

Planking shall be firmly secured to stringers with deformed shank nails to form close butted and battened or tightly overlapped vertical planked panels. All gaps, especially at

bottom, shall not exceed 1% of the total area of the fencing. Ends of panels shall be closed over the gap between stringer and post with return plank material.

741.38 Steel Sidewalk Fencing – Steel sidewalk fence and bicyclist sidewalk fencing shall be installed in accordance with SS Drawings SP741-07.01, SP741-07.02, and SP741-07.03, as applicable.

741.39 Touch-up of Galvanized Coatings – Touch-up treatment for damaged galvanizing of steel posts and braces shall be with two heavy coats of zinc rich paint selected from the Ministry's Recognized Products List under the category of "Additional Paint Coatings – Zinc-Rich Touch-up Paints and Primers.

MEASUREMENT

741.81 Fencing – Fencing will be measured by the LINEAR METRE. Measurements will be made parallel to the ground line of complete fencing, including any tensioning assemblies, but excluding gate openings.

741.82 Gates and Cattleguards – Gates and cattleguards will be measured by the unit for EACH type and size furnished and/or installed complete in place.

PAYMENT

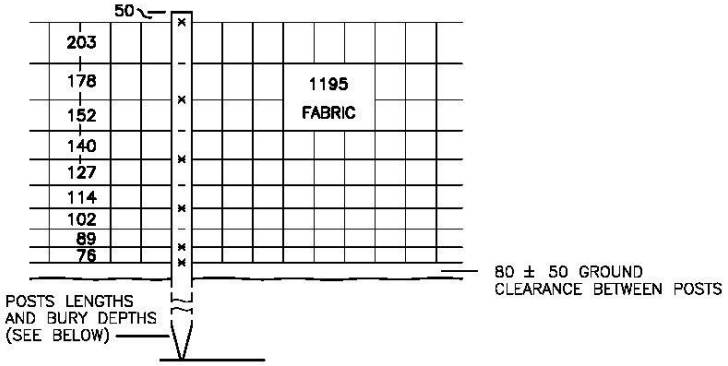
741.91 Fencing – Payment for FENCING will be at the Contract Unit Price per metre of complete fencing, including any tensioning assemblies, but excluding gate openings.

The Contract Unit Price(s) shall be accepted as full compensation for furnishing all material and/or taking delivery of Ministry supplied material; as and where noted, all labour, tools, equipment and incidentals to complete the required installation, including the clearing of any additional right of way, construction of temporary fencing, connection to existing fences and structures, and final clean up; but excluding any work as may be required to be separately paid for, such as the extra cost of extensive fencing on rock at the contract or agreed price for each hole drilled for metal posts.

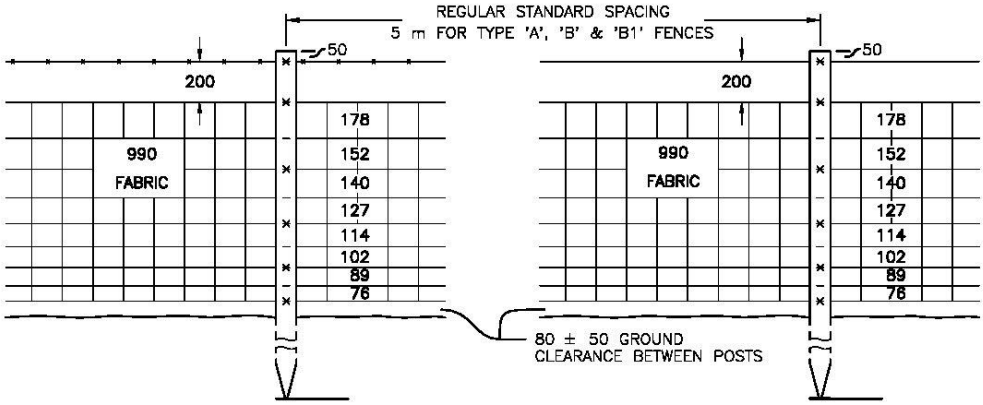
741.92 Gates and Cattleguards – Payment for GATES and CATTLEGUARDS will be at the Contract Unit Price for each type and size furnished and/or installed complete in place.

741.93 Steel Sidewalk Fencing – Payment for steel sidewalk fence and bicyclist sidewalk fencing will be at the Contract Unit Price per metre of complete fencing.

TYPE A, B & B1 STANDARD WIRE FENCE	SP741–01.01



TYPE 'A' FENCE 10476
(RAILWAY RIGHT OF WAY)



* TYPE 'B' FENCE 9396
(HIGHWAY)

TYPE 'B1' FENCE 9396
(HIGHWAY – WILD–LIFE SENSITIVE AREAS)

* USE OF TYPE 'B' FENCE REQUIRES A REVIEW BY THE CHIEF ENVIRONMENTAL OFFICER AND WRITTEN APPROVAL MUST BE RECEIVED PRIOR TO INSTALLATION.

NOTES:

1. FENCE CONSTRUCTION SHALL BE IN ACCORDANCE WITH SECTION 741 OR AS OTHERWISE SPECIFIED OR REQUIRED.
2. PRESSURE TREATED WOOD POST AND BRACE MATERIAL AS SPECIFIED IN SECTION 909
3. DOUBLE STRANDED GALVANIZED BARBED WIRE AS SPECIFIED BY ASTM A 121 AND SUBSECTION 316.06 FOR TYPE 'B' FENCE.
4. SINGLE STRAND HIGH–TENSILE WIRE AS SPECIFIED IN SUBSECTION 316.07 FOR TYPE 'B1' FENCE.
5. WIRE FABRIC AS SPECIFIED BY ASTM A 116 AND SUBSECTION 316.03
6. TYPE 'A' – HEAVY FARM–FIELD GALVANIZED WIRE FABRIC 1195 mm HIGH STYLE 10.47.6.
7. TYPE 'B' – HEAVY FARM–FIELD GALVANIZED WIRE FABRIC 990 mm HIGH STYLE 9.39.6.
8. STAPLES: 45 mm LONG OF 3.75 mm NOMINAL DIAMETER GALVANIZED WIRE TO EVERY LINE OF BARBED AND SMOOTH WIRE AND ALTERNATE LINE WIRE OF FABRIC FENCING AS INDICATED.
9. LINE POSTS: 100 mm MINIMUM DIAMETER, POINTED FOR DRIVING, SET PLUMB AND FIRM.

BURY DEPTH (mm)	POST LENGTH (m)
STANDARD 850	2.2
PEAT SOIL 1050	2.4

10. ROCK, SHALE AND FRIABLE ROCK CONDITIONS WARRANT METAL POSTS – SEE SP741–04.01 FOR MINOR ROCK OUTCROPPINGS AND UNFORESEEN MINIMUM OVERBURDEN OCCURENCES, WIRE TIED MULTIPLE WOOD POST/BRACE ASSEMBLIES MAY BE PERMITTED BY THE MINISTRY REPRESENTATIVE.

NOT TO SCALE
ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

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